

DISSERTATION

HYPERGRAPHS AND THEIR LIE ALGEBRAS

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ABSTRACT

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(need to write abstract)

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Chapter 1

Introduction

In this dissertation we will be exploring a new spectral theory for hypergraphs. Historically, the development of a spectral theory for hypergraphs has followed closely with the spectral theory of graphs, in large part by

1.1 Case Study: Interactions in a High School

To give ourselves an example of a hypergraph which we may study, we present a case study on interactions within a high school. The following data set was retrieved from [2] [13] and [6]. As detailed in [13], a subset of 327 students in a specialized "classes préparatoires" in a French High School were given wearable badges which recorded physical proximity to other students in the study. For this dissertation we will be tracking a random selection of 60 of these students. An edge is made between two students if they were within 1 to 1.5 meters of each other. Mutual interactions in a group, known as cliques, were created by recording interactions within a 20 second interval. If multiple students were in proximity with each other in that 20 second time interval, a simplicial complex was made to record that interaction.

For the purposes of this dissertation we will focus on simplices with three vertices, which for this study would be all 3-faces of the simplicial complexes in the data set. These three faces can be graphed in a 3-way array. We place a nonzero value at position (i,j,k) if students i , j , and k appear in a 3-face in our data set. We can then graph this as a point in $\mathbb{R}^3$, where the coordinates are positive integers. The 60 students we tracked and their 3-cliques are graphed in this manner in Figure 1.1. To the left is the full data set. To the right is the resulting dataset after applying a new algorithm known as stratifications detailed in [9] applied here for the first time to hypergraphs.

For our purposes a stratification is a form of clustering high dimensional data to a lower dimensional surface or structure. This is not a combinatorial optimization problem, nor a method which requires the training of an algorithm. This stratification uniquely identifies features which

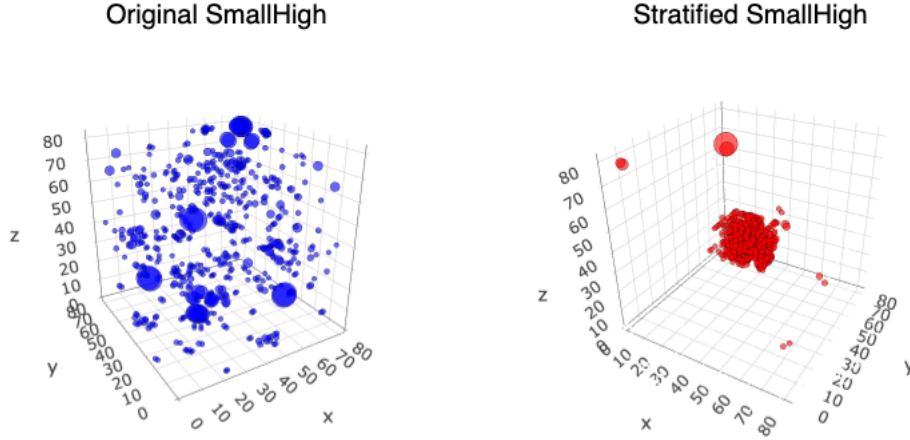


Figure 1.1: High School cliques shown on the left, higher order relations between students shown on the right

were already present in the data set itself. Unlike machine learning methods the features found are reproducible, require no training methods, and are efficiently computed. Furthermore, this method is not simply identifying cliques, which have necessarily been identified in order to store our data. We are also not simply identifying connectivity, which from the data can be found by standard graph theoretic algorithms. This stratification is measuring a higher order of connectivity via linear relations.

For this stratification, a Lie algebra is computed and its elements are used to calculate projections. If the data were truly random we would expect this Lie algebra to be 3 dimensional, as the cliques themselves are three dimensional. However, in our calculations the Lie algebra is at least 10 dimensional (the true dimension of the Lie algebra was not computed due to memory constraints). This means that underlying our data is a more rich structure which can be identified by this algebra. For example, the central cluster in the stratification is indicative of a more connected subset of students that isn't easily observable from the original dataset. This discussion on dimensionality will be continued in Chapter 3. These calculations were our original motivation for using Lie algebras to study hypergraphs. As we will discuss in Chapter 3, it is not immediately obvious that these techniques to analyze tensors should provide information regarding hypergraphs.

Alongside this stratification we computed the barycentric adjacency tensors of this hypergraph, which is a variation on the barycentric subdivision of simplicial complexes. This allows us to subdivide the hypergraph into nested relations of hyperedges. We will develop these tensors in Chapters 2 and 3, and in Section 4.1.1 we will revisit this case study with greater detail. For now, it suffices to know that these tensors count relations between hyperedges that are reminiscent of barycentric subdivisions.

This stratification algorithm and its associated theory had remained unexplored for hypergraphs, and yet it uncovers potential structure which we may further study. For this dissertation, we will be exploring how these algorithms uncover structure within our data, and provide a more detailed description of what this structure means for our hypergraphs.

1.2 Outline

This dissertation is divided into three parts. Chapter 2 will be a review of the necessary algebra we will be using to formalize and study adjacency tensors. In Chapter 3 we will generalize adjacency matrices as adjacency tensors, as well as show how the adjacency tensor can be used to study features of the hypergraph. Finally, in Chapter 4 we will present the main result of this dissertation, which will show how the stratification algorithm is able to identify features in the hypergraph. Moreover, this identification is a bijection between elements of an algebra and hyperedges in the hypergraph.

1.3 The Advantages of Algebra

We will close this chapter with a brief discussion on our motivation to use algebra to study hypergraphs. As inherently combinatorial objects, there are many questions regarding the structure of a hypergraph that can be answered with more combinatorial algorithms and tools. These algorithms tend to be greedy algorithms, which necessarily exploit local features present in a structure. Therefore, the kind of information which is found is very local. We can learn about minimal, maximal, or average vertex degrees, shortest paths between vertices, cliques, etc. In contrast, Algebraic

methods provide a more global insight to data. We can ask about vertex transitivity, or more global symmetries which are present in the data. This provides a fundamental difference in the kinds of questions we will ask and answer regarding hypergraphs in this dissertation.

The work of [15] and [12] defined eigenvalues and eigenvectors on a multilinear product, which has since been used to study eigenvalues of adjacency tensors. [7] utilizes a rescaled adjacency tensor to create an analog for spectral graph theory on hypergraphs, which was used to study bounds on average degrees and coloring of vertices. The Laplacian tensor, constructed by subtracting from the diagonal multiway-array the rescaled adjacency tensor, has been used to study the general connectivity of a hypergraph [8] [11] [14] [1] .

For this dissertation, we will be using an alternative form for a spectral theory of tensors, which has been developed by [9] and [4]. As we will see in Sections 3.2 3.3 this method of defining the spectral properties of tensors more closely aligns with studying matrices and tensors as algebraic objects with a multiplication, as opposed to linear transformations on another space. This allows us to provide a new language for describing eigenvalues that is not constrained to studying inputs to a tensor.

Chapter 2

Tensors

We begin by exploring how tensors may appear in real world data, and what justifications we have for using common tools to study them. The subsequent sections will then further explore linearity and the specific tools we will be using in this thesis to study linear data.

2.1 Distinguishing tensors from grids of numbers

When considering the data that is acquired in the world, it is tempting to run that data through various programs and algorithms in order to extract actionable information. These tools face a great challenge. Barring an egregious error in your data, such as inputting a letter when only integers are expected in the algorithm, these tools will successfully calculate some value.

Consider the following grids of numbers:

	Oil	Wheat	Electronics
Country A	50	-20	10
Country B	-30	40	-15
Country C	-20	-20	5

Figure 2.1: A matrix representing trade

	100m	5k	Half Marathon
Runner A	50	-20	10
Runner B	-30	40	-15
Runner C	-20	-20	5

Figure 2.2: A grid of numbers for runners

In Figure 2.1, we first state that each column represents a different traded good, each row a different country, and each entry the trade surplus or deficit in billions of dollars.

- Scaling a row corresponds to a country increasing either its imports or exports,
- Adding rows corresponds to combining countries into one economic block,
- scaling columns corresponds to the price of the particular good changing, and
- adding columns corresponds to collecting different goods into one class (such as TV's and computers being added to electronics).

	X	Y	Z
A	$4+6z$	$6+7z$	$8+10z$
B	$5+7z$	$4+5z$	$7+10z$
C	$6+11z$	$5+8z$	$6+9z$

Figure 2.3: Supply Chain Tensor

This makes Figure 2.1 a *matrix*. We now take the same grid of numbers, and change their source. In Figure 2.2, the rows represent different runners, the columns represent different races, and the entries represent the difference between the particular runners most recent time in that race vs their personal record. In this context none of the linear operations we would apply to a matrix provide us with *actionable information*. In Chapter 3 we will focus on showing how the adjacency tensors of hypergraphs do provide us with actionable information following specific operations.

As this dissertation studies higher valence tensors, we will give some examples of how such tensors can arise. In Section 2.2 we will provide more specific definitions.

Example 2.1.1. As an example for a higher valence tensor (beyond just a matrix) is with an extension to a standard linear programming problem. Suppose three factories (A, B, and C) produce goods that need to be shipped to three warehouses (X, Y, and Z). Each factory produces a high quality and a low quality version of their product. The cost of each item, including shipping cost, after traveling from each factory to each warehouse is given by Figure 2.3

Where $a + bz$ is a flattening of a multiway array, where a is the cost of the low quality product and b is the cost of the high quality product. We will call this multiway array a *tensor*. Each factory has a limited supply: A can supply 70 units, B can supply 50 units, and C can supply 30 units. Each warehouse has a demand: X needs 40 units, Y needs 60 units, and Z needs 50 units. For the purposes of shipping and storage, each factory and each warehouse considered high and low quality products identical. Therefore, the previous counts of units are the total sum of both the low quality and high quality products. Finally, the total number of high quality goods must be more than twice the total number of low quality goods

$$\begin{aligned}
& \text{Minimize} && 4x_{AX} + 6x_{AY} + 8x_{AZ} + 5x_{BX} + 4x_{BY} + 7x_{BZ} + 6x_{CX} + 5x_{CY} + 6x_{CZ} \\
& && + 6y_{AX} + 7y_{AY} + 10y_{AZ} + 7y_{BX} + 5y_{BY} + 10y_{BZ} + 11y_{CX} + 8y_{CY} + 9y_{CZ} \\
& \text{subject to} && x_{AX} + x_{AY} + x_{AZ} + y_{AX} + y_{AY} + y_{AZ} \leq 70 \\
& && x_{BX} + x_{BY} + x_{BZ} + y_{BX} + y_{BY} + y_{BZ} \leq 50 \\
& && x_{CX} + x_{CY} + x_{CZ} + y_{CX} + y_{CY} + y_{CZ} \leq 30 \\
& && x_{AX} + x_{BX} + x_{CX} + y_{AX} + y_{BX} + y_{CX} = 40 \\
& && x_{AY} + x_{BY} + x_{CY} + y_{AY} + y_{BY} + y_{CY} = 60 \\
& && x_{AZ} + x_{BZ} + x_{CZ} + y_{AZ} + y_{BZ} + y_{CZ} = 50 \\
& && \sum_{i,j} y_{ij} - 2x_{ij} > 0 \\
& && x_{ij} \geq 0 \quad \text{for all } i, j
\end{aligned}$$

Figure 2.4: A linear programming problem presented as a tensor

Let x_{ij} denote the number of low quality units shipped from factory i to warehouse j , and let y_{ij} denotes the number of high quality units shipped from factor i to warehouse j . The linear programming problem is presented in Figure 2.4. As with our example in Figure 2.1, we are justified in treating this data as linear.

In Chapter 3 we will discuss adjacency matrices of graphs, and the additional work necessary to refer to them as *matrices*. The challenge which we are working around is that for a hypergraph, we are representing nodes as vectors. Linearity is then difficult to show, as the addition of vecctors would then correspond to an “addition of vertices,” which has no canonical interpretation.

2.2 Formally defining tensors

As we have seen, simply having data is insufficient to say you are working with linear objects. This is our motivation for giving a purely algebraic definition of tensors. This will follow notation and convention as seen in [4] and [9]. In this dissertation, $\mathbb{F}$ represents a field of characteristic zero, and vectors spaces V will be finite dimensional. In practice, we will chose our fields to be $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{Q}$, but for the purposes of this dissertation we do not need to specify the field.

Definition 2.2.1. Let V_i, W_j be $\mathbb{F}$ vector spaces. A *tensor* is a function

$$\langle t | : V_1 \times \cdots \times V_l \rightarrow V_0,$$

where

$$\langle t | \alpha u_a + \beta v_a, v_{\bar{a}} \rangle = \alpha \langle t | u_a, v_{\bar{a}} \rangle + \beta \langle t | v_a, v_{\bar{a}} \rangle, \quad \forall 1 \leq a \leq l, \quad \forall u_a \in V_a, \quad \forall v_i \in V_i,$$

i.e. the function is *multilinear*.

Throughout this dissertation, $\rightarrow$ will be used when denoting a multilinear map. The property of being multilinear can be thought of via partial evaluations

Definition 2.2.2. A function

$$\langle t | : V_1 \times \cdots \times V_l \rightarrow V_0,$$

is $\mathbb{F}$ *multilinear* if for all $(u_1, \dots, u_l)$, the curried maps

$$d_j : \prod_{i \neq j} V_i \rightarrow (V_j \rightarrow V_0),$$

$$d_j(u_1, \dots, u_{j-1}, u_{j+1}, \dots, u_l) = \langle t | u_1, \dots, u_{j-1}, \cdot, u_{j+1}, \dots, u_l \rangle : V_j \rightarrow V_0$$

are linear maps.

Example 2.2.3. Consider

$$\begin{aligned} \langle t | : \mathbb{R}^2 \times \mathbb{R}^3 &\rightarrow \mathbb{R}^2 \\ \langle t | u, v \rangle &\mapsto \left(u \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} v^T, u \begin{bmatrix} 7 & -3 & 3 \\ -2 & 2 & 4 \end{bmatrix} v^T \right) \end{aligned}$$

As matrix multiplication is linear, this map is linear in both $\mathbb{R}^2$ and $\mathbb{R}^3$.

Often, we define objects as elements of the larger space they inhabit. Within the context of linear algebra, we define vectors as elements of a vector space. In this same vein, we will define tensors as elements of a tensor space.

Definition 2.2.4. Fix $\mathbb{F}$ a field, and finite dimensional $\mathbb{F}$ -vector spaces $V_0, V_1, \dots, V_l$. A *tensor space* is a module T with a $\mathbb{F}$ -linear injection $\langle \cdot | : T \hookrightarrow \text{hom}_{\mathbb{F}}(V_1 \otimes \dots \otimes V_l, V_0)$. A *tensor* t is an element of a tensor space T . $(V_0, \dots, V_l)$ is the *frame* of the tensor space, and V_i is the *i-axis* of the tensor.

It is natural to talk about the *dimension* of a tensor. However, we must use different language as to not be confused by the dimensions of the vector spaces which are the axis of t .

Definition 2.2.5. Let t be a tensor with frame $(V_0, \dots, V_l)$. If $\dim(V_0) \geq 2$, then the *valence* of t is $l + 1$. Otherwise, the valence of t is l .

Example 2.2.6. The tensor from example 2.2.3 has valence 3, and its frame is $(\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^4)$.

For the purposes of this dissertation, we will make the following notational choices. As our vector spaces are finite-dimensional, we may say $V_i \cong \mathbb{F}^{d_i}$. Furthermore, the injection given by $\langle \cdot |$ allows us to interpret tensors t as multilinear functions

$$\begin{aligned} \langle t | : V_1 \times \dots \times V_l &\rightarrow V_0 \\ \langle t | v \rangle &:= \langle t | v_1, \dots, v_l \rangle, \quad v_i \in V_i \end{aligned}$$

An advantage of this interpretation of tensors is that there is a clear distinction between tensors of valence less than or equal to 2, and those of valence greater than or equal to three. We will discuss this in Section 2.4. Our motivation for this discussion will be complex inner product spaces.

Example 2.2.7. The inner product is a tensor. To see this, consider $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined via the standard inner product.

2.3 Frobenius Products and Barycentric Divisor Tensors

In Chapter 3, we will be discussing a spectral theory for hypergraphs. In preparation for that discussion, we will now introduce a new operation on tensors which is reminiscent of the Frobenius product. For matrices, their powers are defined via dot products. Specifically, for a symmetric matrix A , $A^2 = AA^T$ records the information of dot products $a_i \cdot a_j$ for the rows of A . When viewed as a tensor, each e_i is a slice of A along an axis. Therefore, when extending our analogy to tensors, we will be looking at a multiplication between slices of our tensor.

Definition 2.3.1. Let $\langle t | : V^l \rightarrow \mathbb{F}$ be a symmetric tensor. Let $A \subset ([n] \cup *)^m$. An A -slice of $\langle t |$, denoted by $\langle t_A |$, is

$$\begin{aligned} \langle t_A | : V^{l-m} &\rightarrow \mathbb{F} \\ \langle t_A | u &:= \langle t | v_A, u \rangle \\ v_A &:= (e_{a_1}, a_{a_2}, \dots, a_{a_m}), \\ a_i &\in A[i] \end{aligned}$$

Example 2.3.2. Let $\langle t | : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{F}$ be a matrix. The $A = \{2, 3\}$ slice is the slice along the second row and the third column, i.e. $A_{2,3}$. The $A = 2, *$ slice is the slice along the second row and including every column, i.e. the second row $A_{2,*}$.

We can also view $\langle t_A |$ as the partial evaluation of $\langle t |$. As $\langle t |$ is symmetric, we do not need to keep track of specific axis. In the language of matrices, this is equivalent to row i of a symmetric matrix being equal to column i of the same matrix. We will be using an extension of the Frobenius inner product to define our barycentric tensors.

Definition 2.3.3. Let $A, B \in \mathbb{F}^{n_1 \times \dots \times n_l}$ be two multiway arrays. Their *Frobenius inner product* is defined to be

$$\langle A, B \rangle_F := \sum_{a_1, \dots, a_l=1}^{n_1, \dots, n_l} A_{a_1, \dots, a_l} B_{a_1, \dots, a_l}.$$

For multiple $A_1, \dots, A_m \in \mathbb{F}^{n_1 \times \dots \times n_l}$, we can define their *Frobenius product*

$$\langle A_1, \dots, A_m \rangle_F := \sum_{a_1, \dots, a_l=1}^{n_1, \dots, n_l} \prod_{i=1}^m (A_i)_{a_1, \dots, a_l}$$

Utilizing this Frobenius product of tensors, we can define *barycentric tensors*. Note that these will be used in Chapter 3 to discuss barycentric subdivisions.

Definition 2.3.4. Let $\Gamma : V \times \dots \times V \rightarrow \mathbb{F}$ be symmetric. We define the associated *barycentric tensor* as

$$\begin{aligned} \langle t^{(l,1)} | : V^l &\rightarrow \mathbb{F} \\ \langle t^{(l,1)} | e_{a_1}, \dots, e_{a_l} \rangle &= \sum_{i=1}^n \langle \langle t_{\{a_i\}} | \rangle_F \end{aligned}$$

In the 3-valent case, this simplifies to

$$\begin{aligned} \langle t^{(l,1)} | e_i, e_j, e_k \rangle &= \langle \langle t_{i,j} |, \langle t_{i,k} |, \langle t_{j,k} | \rangle_F \\ &= \sum_{l=1}^n \langle t | e_i, e_j, e_l \rangle \langle t | e_i, e_l, e_k \rangle \langle t | e_l, e_j, e_k \rangle \end{aligned}$$

Proposition 2.3.5. Let $\langle t | : \overbrace{V \times \dots \times V}^l \rightarrow \mathbb{F}$ be a symmetric tensor. $\langle t^{(l,1)} |$ is a tensor.

Proof. We will show the proof for the three valent case, as the proof for higher valent tensors follows naturally. Let Γ be a symmetric 3-valent tensor. We will first give a coordinate free definition of $\Gamma^{(3)}$. We define the product

$$[t | u, v, w] := \langle \langle t | u, v, \cdot \rangle, \langle t | u, \cdot, w \rangle, \langle t | \cdot, v, w \rangle, \rangle_F$$

Where each of the terms in the Frobenius product are elements of V^* . As $V^* \cong V$, the Frobenius product is defined naturally. After a change of basis we have

$$[\langle t | Xu, Xv, Xw \rangle = \langle \langle t | Zu, Xv, X(\cdot) \rangle, \langle t | Xu, X(\cdot), Xw \rangle, \langle t | X(\cdot), Xv, Xw \rangle, \rangle_F$$

We can now define an action of $\langle t |$ by a change of basis X by

$$[\langle t |^X | u, v, w \rangle := \langle \langle t | Xu, Xv, X(\cdot) \rangle, \langle t | Xu, X(\cdot), Xw \rangle, \langle t | X(\cdot), Xv, Xw \rangle, \rangle_F$$

This shows us that $(\langle t |^{(3,1)})^X = (\langle t |^X)^{(3,1)}$, i.e. that the barycentric tensor is invariant to change of basis. □

As the barycentric tensor is itself a tensor, we can iterate this product.

Definition 2.3.6. The i^{th} iterated barycentric tensor of a tensor $\langle t |$ is defined to be

$$\begin{aligned} \langle t |^{(l,i)} &:= (\langle t |^{(l,i-1)})^{(l,1)}, \\ \langle t |^{(l,0)} &:= \langle t | \end{aligned}$$

In Section 3.3 we will continue this discussion on barycentric tensors as they relate to adjacency tensors.

2.4 The Centroid

We now return to our discussion on the valence of a tensor. To see where the distinction between 2-valent and 3-valent tensors lies, consider operators on complex inner product spaces V with the standard inner product $\langle x, y \rangle = uv^*$, where u^* is the conjugate transpose. When we apply any operator A to V , we get the following relation.

$$\langle uA, v \rangle = uAv^* = u(vA^*)^* = \langle u, vA^* \rangle.$$

where A^* is the Hermitian adjoint of A . An operator A is *Hermitian* if $A = A^*$. An equivalent definition for a Hermitian operator is $\langle uA, v \rangle = \langle y, vA \rangle$ for all $u, v \in V$. If A and B are two Hermitian operators, then

$$\langle ABu, v \rangle = \langle Bu, Av \rangle = \langle u, BAv \rangle.$$

Unless $AB = BA$, AB is not Hermitian. This is why the space of Hermitian operators is a vector space, as opposed to an associative algebra.

Now, we will extend this analogy to tensors of valence three. Instead of the adjoint and Hermitian operators, we will be using the centroid, which can be thought of as a natural progression from adjoints.

Definition 2.4.1. The *centroid* of a tensor $\langle t | : V_1 \times \cdots \times V_l \rightarrow V_0$ is

$$\text{Cen}(t) := \left\{ A_i \in \prod_{i=0}^l \text{End}(V) \mid \langle t | A_a v_a, v_{\bar{a}} \rangle = A_0 \langle t | v \rangle \right\}$$

Example 2.4.2.

Now, assume we have $(A, A, A), (B, B, B) \in \text{Cen}(\langle t |)$ for a tensor $\langle t | : V \times V \rightarrow V$. Consider (AB, AB, AB) . We have that

$$\begin{aligned} \langle t | ABv_1, v_2 \rangle &= A \langle t | Bv_1, v_2 \rangle = A \langle t | v_1, Bv_2 \rangle = A \langle t | v_1, ABv_2 \rangle \\ \langle t | ABv_1, v_2 \rangle &= \langle t | v_1, ABv_2 \rangle = AB \langle t | v_1, v_2 \rangle \end{aligned}$$

This shuffling is reminiscent of three card monte or towers of Hanoi. With Hermitian operators, we only have two piles to move them around, and so we do not have control over their order. However, once we have a third location for the operator to move to we are able to freely swap positions of the endomorphisms, which tells us the centroid is an algebra.

2.5 Transverse Operators

As we have seen, the centroid is an algebra which we can utilize to study tensors. We now want to see what other algebras exist that can be used to study tensors. In a similar manner to how we study actions on vector spaces as opposed to the vectors themselves, we will discuss operators on tensor spaces.

Definition 2.5.1. Let t be a tensor of the tensor space T , with the frame $(V_0, \dots, V_l)$. *Transverse operators* are elements ω of

$$\Omega := \prod_{i=0}^l \text{End}(V_i)$$

Where $\text{End}(V_i) \cong M_{d_i}(\mathbb{F})$. $\omega = (\omega_0, \omega_1, \dots, \omega_l) \in \Omega$ act on t via the action

$$\langle t^\omega | v \rangle = \omega_0 \langle t | \omega v \rangle = \omega_0 \langle t | \omega_1 v_1, \dots, \omega_l v_l \rangle.$$

For a given tensor and a given operator, we can define a more complex polynomial realtion.

Definition 2.5.2. Let t be a tensor and ω an operator on t . Let $p \in \mathbb{F}[X] := \mathbb{F}[x_0, \dots, x_l]$, where $p(X) = \sum_e \lambda_e X^e$, $e : [[l]] \rightarrow \mathbb{N}_{\geq 0}$, and $X^e := \prod_{i=0}^l x_i^{e(i)}$. Then the action of $p(X)$ on t with a given ω is

$$\langle t | p(\omega) | v \rangle := \sum_e \lambda_e \omega_0^{e(0)} \langle t | \omega_1^{e(1)} v_1, \dots, \omega_l^{e(l)} v_l \rangle$$

Example 2.5.3. Recall that the centroid is defined to be

$$\text{Cen}(t) := \left\{ A_i \in \prod_{i=0}^l \text{End}(V) \mid \langle t | A_a v_a, v_{\bar{a}} \rangle = A_0 \langle t | v \rangle \right\}$$

We may instead express this in the language of transverse operators with the polynomials $p_a(x) = x_0 - x_a$, and operators $(A_0, A_1, \dots, A_l)$.

Example 2.5.4. Let $\langle t | : \mathbb{F}^2 \times \mathbb{F}^3 \times \mathbb{F}^5 \mapsto \mathbb{F}^4$. Set $\omega = (A, B, C, D) \in M_4(\mathbb{F}) \times M_2(\mathbb{F}) \times M_3(\mathbb{F}) \times M_5(\mathbb{F})$. Finally, let $p(X) = 2x_0^2x_1 - 3x_2x_3 + 5x_1^3x_2$. Then

$$\langle t | p(\omega) | v \rangle = 2A^2 \langle t | Bv_1, v_2, v_3 \rangle + (-3) \langle t | v_1, Cv_2, Dv_3 \rangle + 5 \langle t | B^3v_1, Cv_2, v_3 \rangle$$

2.5.1 Derivations

An important class of transverse operators are derivations. AS referenced in Section 1.1, these derivations are what we will be using to construct our tool for studyign hypergraphs, as detailed in Theorem 4.3.3.

Definition 2.5.5. Let $\langle t | \mid V_l \times \cdots \times V_1 \mapsto V_0$ be a tensor. Its *derivation algebra* $\text{Der}(\langle t |)$ is

$$\text{Der}(\langle t |) := \left\{ D \in \text{End}(V_0) \times \prod_{i=1}^l \text{End}(V_i) \mid \sum \langle t | D_\alpha v_\alpha, v_{\bar{\alpha}} \rangle = D_0 \langle t | v \rangle \right\}$$

Example 2.5.6. Given the tensor

$$\begin{aligned} \langle M | : \mathbb{R}^2 \times \mathbb{R}^3 \mapsto \mathbb{R}^1 \\ \langle u | M | v \rangle \mapsto u \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} v^T, \end{aligned}$$

the derivation algebra $\mathcal{D}$ of Γ is the set of triples $(D_0, D_1, D_2) \in M_1(\mathbb{R}) \times M_2(\mathbb{R}) \times M_3(\mathbb{R})$ that satisfy Leibniz on a basis, i.e.

$$D_0 \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} = D_1 \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} D_2^T$$

As its name suggests, the derivation algebra is indeed an algebra. However, it is a non-associative algebra, namely a Lie algebra.

Definition 2.5.7. A *Lie Algebra* is an algebra $\mathfrak{g}$ whose multiplication satisfies the following conditions

- $x^2 = 0$ (Alternating), and
- $(xy)z + (yz)x + (zx)y = 0$ (Jacobi Identity)

Example 2.5.8. Let $GL(n, \mathbb{C})$ be the vector space of invertible $n \times n$ matrices over $\mathbb{C}$. Define the product of two matrices X and Y via the commutator $[X, Y] = XY - YX$. $GL(n, \mathbb{C})$ with the as its multiplication a Lie algebra.

Proposition 2.5.9. Let $\langle t|$ be a tensor. $\text{Der}(\langle t|)$ is a Lie Algebra.

Proof. We take the set of tuples of matrices $\text{Der}(\langle t|)$, and define addition component wise, scaling as distributive, and the multiplication via component wise commutators. It can then be verified that these operations are closed, and they satisfy all properties to make $\text{Der}(\langle t|)$ a Lie algebra. $\square$

Not all Lie algebras are known to occur as lie algebra of tensor, yet it is known that every simple lie algebra is known to be the derivation of a tensor. We will be interested in the simple Lie algebras that occur when studying our hypergraphs.

As we will see in Chapter 4, and specifically Section 4.2.1, the fact that $\text{Der}(\langle t|)$ is a Lie algebra will allow us to make more informed choices for generators. It is from these generators that we construct our different cases in Theorem 4.3.3, and why we are able to check finitely many cases.

2.6 TIZ Correspondence

This section goes through one of the main results of [9]. We first fix a tensor space $T := \mathbb{F}^{d_0} \otimes \dots \otimes \mathbb{F}^{d_l}$. We then have transverse operators $\Omega := M_{d_0}(\mathbb{F}) \times \dots \times M_{d_l}(\mathbb{F})$ and the ring of polynomials $\mathbb{F}[X] := \mathbb{F}[x_0, \dots, x_l]$. For a fixed set of tensors $S \subset T$ and operators $\Delta \subset \Omega$, define

$$\mathbf{I}(S, \Delta) \left\{ p(X) := \sum_e \lambda_e X^e \mid \forall t \in S, \forall \omega \in \Delta, \langle t| p(\omega) = 0 \right\}$$

where we say $\langle t | p(\omega) = 0$ if and only if $\langle t | p(\omega) | v \rangle = 0$ for all $v \in \mathbb{F}^{d_1} \otimes \cdots \otimes \mathbb{F}^{d_l}$, $\langle t | p(\omega) | v \rangle = 0$. Our goal is to look at the zero loci of $\mathbf{I}(S, \Delta)$. Next, with $P \subset \mathbb{F}[X]$ we define

$$\mathbf{N}(P, \Delta) := \{t \in T \mid P \subset I(t, \Delta)\}.$$

This is the set of all tensors for which the given polynomials in P lie within $\mathbf{I}(\{t\}, \Delta)$. It can be thought of as tensors satisfying the relation $\langle t | p(\omega) = 0$. Finally, we define a set of transverse operators.

$$\mathbf{Z}(S, P) := \{\omega \in \Omega \mid P \subset \mathbf{I}(S, \omega)\}$$

Similarly to $\mathbf{N}(P, \Delta)$, $\mathbf{Z}(S, P)$ is the set of operators satisfying the relation $\langle t | p(\omega) = 0$. The following examples give further insight into these three sets.

Example 2.6.1. Let $l = 0$, $T = \mathbb{F}^{d_0}$, and $\Delta = \{\omega\}$. Then $\mathbf{I}(T, \Delta) = (\text{minpoly}_\omega(x) \subset \mathbb{F}[x])$, and then the zero loci of $\mathbf{I}(T, \Delta)$ are the zeros of the ideal generated by the minimal polynomial of ω , i.e. the eigenvalues of ω .

Example 2.6.2. Let $l = 0$, $P = (x - \lambda)$, and fix $\omega \in M_{d_0}(\mathbb{F})$. Then $\mathbf{N}((x - \lambda), \omega) = \{t \in K^{d_0} \mid \omega t = \lambda t\}$ is the eigenspace of ω .

Example 2.6.3. Let $l = 0$, $T = \mathbb{F}^{d_0}$, and $P = (p(x))$. Then $\mathbf{Z}(S, P)$ is the set of matrices ω for which $p(\omega) = 0$, i.e. the matrices which have at least one eigenvalue which is a root of $p(x)$.

The following is the main result of [9]

Theorem 2.6.4. [First-Maglione-Wilson] TIZ Correspondence Let $P \subset \mathbb{F}[X]$, $S \subset \mathbb{F}^{d_0} \otimes \cdots \otimes \mathbb{F}^{d_l}$, and $\Delta \subset \Omega$ Then $\mathbf{I}(S, \Delta)$ is an ideal of $\mathbb{F}[X]$ $\mathbf{N}(P, \Delta)$ is a subspace of T , and $\mathbf{Z}(S, P)$ is a closed subscheme of Ω . Furthermore, the following Galoid connection holds

$$S \subset \mathbf{N}(P, \Delta) \iff P \subset \mathbf{I}(S, \Delta) \iff \Delta \subset \mathbf{Z}(S, P)$$

2.6.1 Densors

Once we have higher valence tensors, we will utilize a different algebra to construct our decompositions.

From the language of Theorem 2.6.4, we have an equivalent definition of the derivation algebra. Recall definition 2.5.5 which says that for a tensor $\langle t|$, the derivations are tuples of endomorphisms D such that $\sum \langle t|D_\alpha v_\alpha, v_{\bar{\alpha}} \rangle = D_0 \langle t|v \rangle$. We may now view this in the context of TIZ.

By construction, $\delta \in \text{Der}(\langle t|)$ if and only if

$$\delta_0 \langle t|v \rangle - \sum_{a=1}^l \langle t|\delta_a v_a, v_{\bar{a}} \rangle = 0$$

Let $\langle t|$ be a tensor, and set $p = x_0 - \sum_{i=1}^l x_i \in \mathbb{F}[X]$. The derivations of $\langle t|$, denoted by $\text{Der}(\langle t|)$, is the set $\mathbf{Z}(\langle t|, p)$.

This is equivalent to

Let $\langle t|$ be a tensor, and set $p = x_0 - \sum_{i=1}^l x_i \in \mathbb{F}[X]$. The derivations of $\langle t|$, denoted by $\text{Der}(\langle t|)$, is the set $\mathbf{Z}(\langle t|, p)$.

which aligns with our previous characterization of derivations.

Proposition 2.6.5. Let $\langle t|$ be a tensor, and set $p = x_0 - \sum_{i=1}^l x_i \in \mathbb{F}[X]$. The set of derivations $\text{Der}(\langle t|)$ with the following operations

1. $\alpha(\delta_0, \dots, \delta_l) = (\alpha\delta_0, \dots, \alpha\delta_l)$,
2. $\delta + \gamma = (\delta_0 + \gamma_0, \dots, \delta_l + \gamma_l)$, and
3. $\delta \bullet \gamma = ([\delta_0, \gamma_0], \dots, [\delta_l, \gamma_l])$

is a Lie algebra, where $[\delta_i, \gamma_i] = \delta_i\gamma_i - \gamma_i\delta_i$ is the standard commutator within the space of Endomorphisms.

Proof. 1 and 2 follow from $\langle t|$ being multilinear, and derivations being defined as a relation on a summation. 3 can be verified by direct computation. □

2.7 Chisels

When constructing the densor space, we are allowed to make choices for our polynomial p . Each choice of p , called a *chisel* provides a new space which we may sample. We use these chisels to decompose our tensors.

Definition 2.7.1. Let $\langle t | : V_1 \times \cdots \times V_l \rightarrow \mathbb{F}$ be a tensor. A *sparcity pattern* is a decomposition of the V_i such that

- $V_i = \bigoplus W_{i,i}$, where $W_{i,i}$ is a $\mathbb{F}$ vector space,
- After relabeling, $\langle t | W_{1,1}, W_{2,1}, W_{\overline{12},a} \rangle = 0$

As derivations algebras are Lie algebras, they carry with them their own decompositions.

Theorem 2.7.2. Levi Let $\mathfrak{g}$ be a Lie algebra. $\mathfrak{g}$ decomposes as $\mathfrak{S} \oplus \mathfrak{R}$, where $\mathfrak{S}$ is semisimple and $\mathfrak{R}$ is radical.

This Levi decomposition is where we get our projections to define our chisels. $\mathfrak{S} = \bigoplus \mathfrak{g}_i$, where $\mathfrak{g}_i$ are simple Lie algebras. We can then take elements from each simple Lie algebra and project along those elements. The images in Figure 1.1 were created with this chisel algorithm. At the moment, the main implementation of this algorithm will select random elements from each simple Lie algebra. With Theorem 4.3.3, we have a more systematic and informative way of selecting derivations.

2.8 Symmetry in tensors

We will be studying tensors that arise from hypergraphs, and these adjacency tensors have symmetries which are not present in all hypergraphs. In this section, we will formally define these symmetries.

First, we will briefly discuss the matter of covariance and contravariance. In general, a tensor is expressed as

$$\langle t | : V_1 \times \cdots \times V_{l-1} \rightarrow V_l,$$

$$V_i \cong \mathbb{F}^{d_i}.$$

We can construct isomorphism from $\langle t|$ and $\langle s|$, where

$$\langle s| : V_1 \times \cdots \times V_{l-1} \times V_l \rightarrow \mathbb{F}$$

We will first look at this isomorphism through a smaller example.

Example 2.8.1. Let

$$\begin{aligned} \langle t| : \mathbb{R}^2 &\rightarrow (\mathbb{R}^3)^* \\ u &\mapsto u^T \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \langle s| : \mathbb{R}^2 \times \mathbb{R}^3 &\rightarrow \mathbb{R} \\ \langle s| u, v &\mapsto \langle t| u \rangle v = \left(u^T \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} \right) v \end{aligned}$$

As $\mathbb{R}^3$ is finite dimensional, there is a natural isomorphism $\mathbb{R}^3 \cong (\mathbb{R}^3)^*$, giving us a natural isomorphism between $\langle t|$ and $\langle s|$.

The isomorphism between $\langle t|$ and $\langle s|$ in general is made through partial evaluations and the isomorphism of $\mathbb{F}^n \cong (\mathbb{F}^n)^*$. Furthermore, this isomorphism will allow us to choose whether to use $\langle t|$ or $\langle s|$. This will come up as we discuss symmetry in transverse operators.

Definition 2.8.2. A tensor $\langle t| : V_1 \times \cdots \times V_l \rightarrow \mathbb{F}$ is *symmetric* if

1. $V_i = V_j$ for all i, j , and
2. $\langle t|v \rangle = \langle t|v_{\sigma(1)}, \dots, v_{\sigma(l)} \rangle$ for all σ in the symmetric group S_l .

In other words, a symmetric tensor is one in which all of its inputs come from the same vector space, and permuting the inputs does not change the final calculation. This is the analog of symmetric tensors, and is important for studying adjacency tensors as we are not assuming an ordering on vertices.

Example 2.8.3. Let $\langle t| : V_1 \times V_2 \rightarrow \mathbb{F}$. Then $\langle t|$ is symmetric as a tensor if and only if it is symmetric as a matrix.

When a tensor is defined via a higher valence array, we get an equivalent definition.

Definition 2.8.4. Let $A : [n] \times \cdots [n] \rightarrow \mathbb{F}$ be the l -valent array representing the tensor $\langle t|$. Then $\langle t|$ is *symmetric* if

$$A(x_1, x_2, \dots, x_l) = A(x_{\sigma(1)}, \dots, x_{\sigma(l)})$$

For all permutations σ in S_l .

Given a symmetric tensor $\langle t| : V \times \cdots \times V \rightarrow \mathbb{F}$, we will also refer to the following related tensor as *symmetric*

$$\begin{aligned} \langle s| : V^{l-1} &\rightarrow V^*, \\ \langle s|v_1, \dots, v_{l-1}\rangle &:= \langle t|v_1, \dots, v_{l-1}, \cdot \rangle, \end{aligned}$$

where $\langle s|$ is the partial evaluation of $\langle t|$ with the first $l - 1$ inputs being $(v_1, \dots, v_{l-1})$. As before, $V^* \cong V$, so we may write $\langle s| : V^{l-1} \rightarrow V$. We will often refer to $\langle t|$ and $\langle s|$ as representing the same data.

Another aspect of symmetry is in the operators acting on a tensor.

Definition 2.8.5. Let $\omega \in \prod_{i=0}^l \text{End}(V)$ be a transverse operator on a symmetric tensor $\langle t| : V^l \rightarrow \mathbb{F}$. ω is a *symmetric operator* if $\omega_i = \omega_j$ for all $0 \leq i, j \leq l$.

When $\omega_0 \in \text{Aut}(V)$, then the symmetric operator ω is applying a change of basis to the vector space V on which $\langle t|$ acts. These symmetric operators are what define the cases of Theorem 4.3.3

Chapter 3

Hypergraphs

In this chapter, we will formally define hypergraphs, and establish a spectral theory to study adjacency matrices and tensors. From here on, we will also make a notational change to our tensors and tensor evaluations. We will write

$$\Gamma(u, v, w) := \langle t | u, v, w \rangle.$$

This is to more closely align our notation to one of an inner product (u, v) .

3.1 What is a Hypergraph

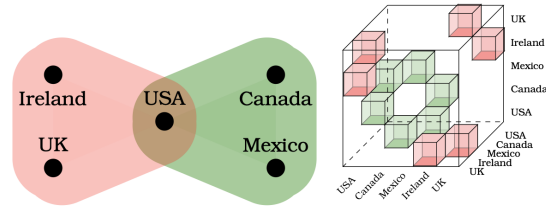
In this section, we will establish the notation we will be using to describe hypergraphs. This will follow similar notation to [3].

Definition 3.1.1. A *hyperedge* $h \subset V$ is a subset of vertices. A *hypergraph* $\mathcal{H}$ is a collection of hyperedges.

Example 3.1.2. A graph is an example of a hypergraph where all hyperedges have either two elements (edges) or one element (isolated vertices).

Example 3.1.3. We can look at trade agreements as an example of hypergraphs. in Figure 3.1 each hyperedge represents a trade agreement, while each vertex represents a distinct nation.

We provide three different visualizations of this hypergraph. First, we have a more pictorial representation where each filled in shape (here triangles) represents a trade agreement, and each vertex is a different nation. We then represent this data in an adjacency multiway array. Finally, we use an incidence matrix, where each row represents a different nation, each column a different trade agreement, and we record which nations are included in each trade agreement.



	Red	Green
USA	1	1
Canada	0	1
Ireland	1	0
UK	1	0
Mexico	0	1

Figure 3.1: Trade Agreement Hypergraph

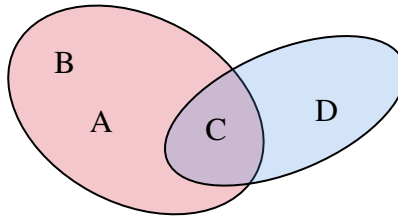


Figure 3.2: Example non uniform hypergraph

In Example 3.1.3, one of the ways that we chose to represent a hypergraph was with a multiway array. It is not always the case that we can do this. Consider the hypergraph with incidence matrix

	Edge 1	Edge 2
Vertex A	1	0
Vertex B	1	
Vertex C	1	1
Vertex D	0	1

This hypergraph can be visualized in Figure 3.2. When making a multiway array to encode the data for a hypergraph, if not all hyperedges have the same number of vertices, we cannot record all hyperedges. For this reason we work with *uniform* hypergraphs.

Definition 3.1.4. Let $\mathcal{H}$ be a hypergraph. $\mathcal{H}$ is l -uniform if $|h| = l$ for all $h \in \mathcal{H}$.

Proposition 3.1.5. There exists a bijective map from the set of all hypergraphs to the set of all labeled uniform hyeprgraphs with at most two labels.

Proof. Let $\mathcal{H}$ be an arbitrary hyeprgraph. Define $k = \max_{h \in \mathcal{H}} |h|$. For each hyperedge e which contains $j < k$ vertices, adjoin to it $k - j$ new vertices $(w_e)_1, \dots, (w_e)_{k-j}$, so that

$$\{v_1, v_2, \dots, v_k\} \mapsto \{v_1, v_2, \dots, v_k, (w_e)_1, (w_e)_2, \dots, (w_e)_{k-j}\}.$$

For each hyperedge we use distinct w_i 's to construct our new hypergraph. For our labeling of $\overline{\mathcal{H}}$, we label our original vertices A , and the newly adjoined vertices B .

For any labeled uniform hypergraph with two labels, we chose one of the labels, and for each hyepredge remove all vertices with that label. The resulting object is an unlabeled hypergraph. $\square$

A different approach is to add a singular point at infinity to “complete” each hyperedge, as was done by [16] when using an embedding of the hypergraph into a lower dimensional space to determine the closeness of communities. In this paper, we are adding multiple distinct points so that we do not accidentally further relate hyperedges beyond the connections found in the original hypergraph. With Lemma 3.1.5, we can regard all hypergraphs as uniform, as long as we retain the information of which vertices were added after applying the bijection.

We can now formally define the *adjacency tensor*

Definition 3.1.6. For a given l -uniform hypergraph $(V, \mathcal{H})$, the associated *adjacency tensor* is the multilinear map

$$\Gamma : \overbrace{V \times \dots \times V}^l \rightarrow \{0, 1\},$$

$$\Gamma(x_1, x_2, \dots, x_l) := \begin{cases} 1 & \{x_1, x_2, \dots, x_l\} \in \mathcal{H} \\ 0 & \text{else} \end{cases},$$

$$x_i = e_j$$

In Section 3.2 we will be showing why we may refer to this as a tensor.

3.2 Spectral Theory of Hypergraphs

In this section, we will justify why we are able to call the multiway array that records adjacency of a hypergraph a *tensor*. We begin with the adjacency matrix of a graph, as it is a well studied and commonly known example. We will assume that all graphs are undirected, and thus their adjacency matrix is symmetric

3.2.1 Adjacency matrices

Given a graph $G = (V, E)$ with n vertices, we construct the adjacency matrix $A \in \mathbb{C}^{n \times n}$ as follows

$$(A)_{ij} := \begin{cases} 1 & (i, j) \in E \\ 0 & \text{else} \end{cases}$$

If A were a matrix we would be justified in applying matrix operations to this. However, we do not apply such operations, as row and column operations do not hold meaning in regards to the underlying graph. What we are left with is taking powers of the adjacency matrix. A^2 records the number of walks of length 2 from i to j , given by $(A^2)_{ij}$. In general, $(A^k)_{ij}$ is the number of walks of length k from i to j . In this sense, it is not row operations but rather dot products that inform us of the graph. For two square matrices, $(AB^T)_{ij} = a_i \cdot b_j$, where a_i is row i of A , b_j is row j of B , and $\cdot$ is the standard dot product. For the adjacency matrix of a graph,

$$(A^2)_{ij} = (AA^T)_{ij} = a_i \cdot a_j$$

Returning to row i of A , this has a nonzero value at position k if and only if $(i, k) \in E$. This means that for the dot product $a_i \cdot a_j = \sum_{k=1}^n (a_i)_k (a_j)_k$, nonzero terms only appear when both $(i, k) \in E$ and $(j, k) \in E$, i.e. vertices i and j share a neighbor. Since the number of walks between vertices

i and j are exactly equal to the number of shared neighbors between i and j , this validates our interpretation of A^2 as counting the number of length 2 walks between vertices. Continuing with this change of perspective, A^3 counts the number of paths of length 3 from vertex i to vertex j

$$(A^2 A)_{ij} = \sum_{k=1}^n ((A^2)_{i*})_k (a_j)_k$$

We already know that $(A^2)_{ik}$ counts the number of neighbors shared between i and k . When multiplying $(A^2)_{i*} \cdot a_j$, we get a nonzero value only for $(a_j)_k \neq 0$, i.e. at positions corresponding to neighbors of vertex j . Therefore, $(A^2)_{i*} \cdot a_j$ has contributing nonzero values only when a neighbor of vertex j is equal to a vertex which is of path length 2 from vertex i , or equivalently the neighbor of a neighbor of vertex i . Inductively, A^k can be seen as counting these “neighbor of a neighbor” or “friend of a friend” relations between vertices (where repetitions are allowed).

We now do a similar computation for adjacency tensors. We will first look at defining a typical inner product for slices of an adjacency tensor which is a vector, and then see how this extends to inner products on higher valent slices. Let Γ be the adjacency tensor for $\mathcal{H} = \{\{1, 2, 3\}, \{3, 4, 5\}\}$. Recall Definitions 2.3.3 and 2.3.4 of the Frobenius product. Consider the two slices $\Gamma_{1,2}$ and $\langle s|_{4,5}$. When viewed as multiway arrays, $\Gamma_{1,2} \cong (0, 0, 1, 0, 0, 0)$ and $\Gamma_{3,4} \cong (0, 0, 1, 0, 0, 0)$. The Frobenius inner product of these two slices is equal to 1, which can be restated as “there is exactly one vertex which is in a hyperedge with both $\{1, 2\}$ and $\{4, 5\}$.” This is our first sign that we are justified in calling adjacency multiway arrays *tensors*.

3.3 Spectral Barycenters

Continuing our study of the spectral properties of adjacency tensors from section 3.2, we recall Definition 2.3.4 for the barycentric tensor. When the tensor arises as an adjacency tensor of a hypergraph, we will instead refer to the *barycentric adjacency tensor*. As we have named these barycentric, we will show what geometric feature they count. Let Γ be the adjacency tensor of

a 3-uniform hypergraph. If we consider each term $\Gamma_{ijk}^{(3,1)}$, we get a nonzero value if and only if $\{i, j, l\}, \{i, l, k\}, \{l, j, k\} \in \mathcal{H}$ for some $l \in V$. This can be viewed as the barycentric subdivision of the simplex $\{i, j, k\}$ with a fourth point l . A key distinction here is that $\{i, j, k\}$ itself does not necessarily need to be a hyperedge in $\mathcal{H}$. This allows the barycentric adjacency tensor to find relations that may be missed from a traditional barycentric subdivision. From here, $(\Gamma^{(l,1)})_{a_1 \dots a_l}$ counts the number of vertices which can be used to identify a barycentric subdivision of the simplex $\{a_1, \dots, a_l\}$. By Lemma 2.3.5, this is itself a tensor.

As with characteristic polynomials of a matrix, we can define a characteristic polynomial for these barycentric tensors.

Theorem 3.3.1. Let Γ be the adjacency tensor of an l -uniform hypergraph $\mathcal{H}$, and let $\Gamma^{(l,i)}$ be the i^{th} iterated barycentric tensor of Γ . Then there exists a unique minimal finite k such that

$$\alpha_0 \Gamma + \sum_{i=1}^k \alpha_i \Gamma^{(l,i)} = 0$$

We refer to this as the *barycentric characteristic polynomial*.

Proof. Let $\Gamma^{(l,i)}$ be the i^{th} iterated barycentric tensor of Γ . Each $\Gamma^{(l,i)}$ and Γ is an element of the tensor space $T = \overbrace{\mathbb{F}^n \otimes \dots \otimes \mathbb{F}^n}^l$. T is finite dimensional with a basis $\{e_{i_1} \otimes \dots \otimes e_{i_l}\}$. Once we have calculated $\Gamma^{(l,n^l)}$, we have a list of $n^l + 1$ elements in an n^l -dimensional space, and so there must exist a linear relationship within the set of tensors. $\square$

We note that this linear relation is reminiscent of a characteristic polynomial. For the adjacency matrix of a graph A , Theorem 3.3.1 refers simply to the characteristic polynomial. Here, $\Gamma^{(2,i)}$ is simply A^i . Our interpretation of A^i is simplified in that matrices are 2-valent. A^i is the composition of linear transformations, and so we can clearly define $A^i u = \lambda u$ to define eigenvalues, and it can be shown that these eigenvalues are the roots of the characteristic polynomial. There is an immediate challenge once the valence of our tensor is greater than 2. A higher valence tensor $\langle t |$ must have either at least two inputs, or at least two outputs. From here there is no

longer a canonical choice for an iteration. The work of [15] and [12] made the choice to consider $\langle t|u, u, \dots, u \rangle = \lambda^l u$ to define eigenvalues, however this restricts us to considering our tensors as functions with a single input $F(u) := \langle t|u, u, \dots, u \rangle$. The goal of this dissertation is to consider adjacency tensors as the algebraic objects they are, and so these additional features lead us to future avenues of research.

Chapter 4

Derivations for Hypergraph Features

In this Chapter, we will be proving Theorem 4.3.3. This theory builds on work done by [9] and [4] in studying derivation algebras. Furthermore, we will be looking at 3-uniform hypergraphs in this dissertation, but this theory can be expanded to higher order hypergraphs.

4.1 Motivating Derivations

Consider the tensor.

$$\Gamma = \left[\begin{array}{c} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ \{1\} \end{array}, \begin{array}{c} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \{x\} \end{array} \right]$$

The labels 1 and x are used to keep track of the slices of our $2 \times 2 \times 2$ tensor. We could have stored this information in a multiplication table.

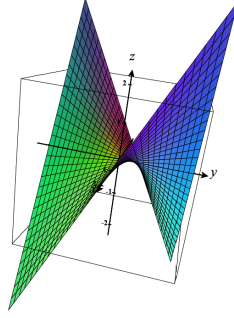
*	1	x
1	1	x
x	x	-1

This translation happened by viewing 1 and x as the vectors $(1, 0)$ and $(0, 1)$. This multiplication table is really expressing $\mathbb{R}[x]/(x^2 + 1) = \mathbb{C}$. This tells us that our tensor was really holding onto the information of the complex numbers, which we were able to deduce by viewing it as a multiplication table. This is our motivation for studying products.

Now that we have products, lets look at the most simple product. Take

$$x = f(x, y) = xy.$$

As a analysis, we would graph this function.



We can then study the curvature of This function. This is done by computing derivatives. Since we have a product, we need ot work with the product rule of derivatives

$$\frac{d}{dx}(f(x)g(x)) = \left(\frac{d}{dx}f(x)\right)g(x) + f(x)\left(\frac{d}{dx}g(x)\right)$$

Let us now rewrite this rule. Define $f(x)g(x) := \Gamma(f(x), g(x))$, and $D := \frac{d}{dx}$. Then the product rule is

$$D\Gamma(f(x), g(x)) = \Gamma(Df(x), g(x)) + \Gamma(f(x), Dg(x))$$

This is our motivation for looking at derivations.

Definition 4.1.1. Let $\Gamma \mid V_l \times \cdots \times V_1 \rightarrow V_0$ be a tensor. Its *derivation algebra* $\text{Der}(\Gamma)$ is

$$\text{Der}(\Gamma) := \left\{ D \in \text{End}(V_0) \times \prod_{i=1}^l \text{End}(V_i) \mid \sum \langle \Gamma \mid D_\alpha v_\alpha, v_{\bar{\alpha}} \rangle = D_0 \langle \Gamma \mid v \rangle \right\}$$

Example 4.1.2. Given the tensor

$$\langle t \mid : \mathbb{R}^2 \times \mathbb{R}^3 \rightarrow \mathbb{R}^1,$$

$$\langle t \mid u, v \rangle := u \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} v^T$$

the derivation algebra $\mathcal{D}$ of $\langle t|$ is the set of triples $(D_0, D_1, D_2) \in M_1(\mathbb{R}) \times M_2(\mathbb{R}) \times M_3(\mathbb{R})$ that satisfy Leibniz on a basis, i.e.

$$\begin{bmatrix} D_0(1) & D_0(2) & D_0(3) \\ D_0(-2) & D_0(1) & D_0(4) \end{bmatrix} = D_1 \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ -2 & 1 & 4 \end{bmatrix} D_2^T$$

In practice, the derivation algebra is computed by solving large systems of linear equations. We have $\prod_{i=0}^l (\dim(V_i))^2$ variables and a system of $\prod_{i=1}^l \dim(V_i)$ equations. We can give a minimal dimension of the derivation algebra.

Lemma 4.1.3. Let $\langle t| : V_1 \times \cdots \times V_l \rightarrow \mathbb{F}$ be a tensor. Then $\dim(\text{Der}(\langle t|)) \geq l$.

Proof. We will show that there exists at least l linearly independent derivations of $\langle t|$. As $\langle t|$ is multilinear we have that

$$\langle t|\alpha u_a, u_{\bar{a}}\rangle = \alpha \langle t|u\rangle.$$

This can be rewritten as

$$\langle t|(\alpha I)u_a, u_{\bar{a}}\rangle = \alpha \langle t|u\rangle.$$

This allows us to generate l derivation

$$\begin{aligned} &(\alpha, \alpha I, 0, \dots, 0), \\ &(\alpha, 0, \alpha I, \dots, 0), \\ &\vdots \\ &(\alpha, 0, \dots, \alpha I). \end{aligned}$$

These are clearly linear independent, and as such the dimension of $\text{Der}(\langle t|)$ is at least l . □

It is important to note that the derivation algebra is a non-associative algebra. Indeed, it is a *Lie algebra*

Definition 4.1.4. A *Lie Algebra* is an algebra $\mathfrak{g}$ whose multiplication satisfies the following conditions

- $x^2 = 0$ (Alternating), and
- $(xy)z(yz)x + (zx)y = 0$ (Jacobi Identity)

Example 4.1.5. Let $GL(n, \mathbb{C})$ be the vector space of invertible $n \times n$ matrices over $\mathbb{C}$. Define the product of two matrices X and Y via the commutator $[X, Y] = XY - YX$. $GL(n, \mathbb{C})$ with the as its multiplication a Lie algebra.

Lemma 4.1.6. $Der(\Gamma)$ is a Lie Algebra.

Proof. We take the set of tuples of matrices $Der(\Gamma)$, and define multiplication component wise and the multiplication via component wise commutators. It can then be verified that these operations are closed, and they satisfy all properties to make $Der(\Gamma)$ a Lie algebra. $\square$

4.1.1 Revisiting Our Case Study

Before we begin the proof of Theorem 4.3.3, we will revisit the case study we began in Section 1.1. Recall that the vertices of our hypergraph were studetns, and hyperedges were created by constructing cliques of students that were in close proximity to one another ina 20 second time period. This information was stored in an adjacency tensor Γ . The derivation algebra $Der(\Gamma)$ is then computed. As detailed in Section 4.1 we solve a system of linear equations to get our solution for the full derivation algebra. In the implimentation of this chiseling algorithm we used, a random element (D_0, D_1, D_2, D_3) . The projection is then as follows:

$$V \rightarrow D_1 V,$$

$$V \rightarrow D_2V,$$

$$V \rightarrow D_2V,$$

$$\mathbb{F} \rightarrow D_0\mathbb{F}.$$

Once this is done, we reconstruct the tensor as $\hat{\Gamma} : D_1V \times D_2V \times D_3V \rightarrow D_0\mathbb{F}$. This new tensor $\hat{\Gamma}$ is the second image in Figure 1.1.

After we computed the derivation algebra of the full data set, the dimension of this Lie algebra was (computation needed). The minimal dimension of this Lie algebra is three, and the derivation algebra for a random 3-valent tensor will be three or close to three. The fact that the derivation algebra is much larger than expected tells us that there are additional symmetries present in the adjacency tensor which are being found by the endomorphisms in the derivation algebra.

4.2 Generating Lie Algebras

As our goal is to use the Lie algebra of derivations arising from the adjacency tensor of a hypergraph, we will take a moment to study these derivations as Lie algebras. This will require us to establish background work on Lie algebras, and will allow us to focus our study into matrices with support less than four.

To begin, we assert that we will consider our derivation algebras under the natural basis. By construction, we are mapping vertices in our hypergraph to the standard basis vectors. From here, when we construct our adjacency tensor we construct it to act on tuples of these standard basis vectors. Finally, the derivation algebra is constructed as tuples of matrices which act on these standard basis vectors. While further work may be done to study other representations of the Lie algebra of derivations of the adjacency tensor, that is beyond the scope of this dissertation.

By focusing on the natural representation, we gain beneficial restrictions to basis elements.

Lemma 4.2.1 (Chevalley [5]). Let $\mathfrak{L}$ be a finite dimensional simple Lie algebra. There exists a basis $\mathcal{U}$ of $\mathfrak{L}$ such that for all $u \in \mathcal{U}$, the support of u is less than or equal to three.

The proof for this lemma will be modeled from [10].

Proof. We will consider the generators of the simple Lie algebras.

$-A_n$, or $\mathfrak{sl}_n$. The elements of $\mathfrak{sl}_n$ can be identified with $n \times n$ matrices of trace 0. We denote the standard basis of $n \times n$ matrices by E_{ij} , where

$$E_{ij}E_{km} = \delta_{jk}E_{im},$$

$$\sum E_{ii} = I_n.$$

We can express a basis for $\mathfrak{sl}_n$ as

$$H_k = E_{kk} - E_{nn}, \quad k < n,$$

$$E_{ij}, \quad i \neq j, \quad 1 \leq i, j \leq n$$

$-B_n$, or $\mathfrak{so}_{2n+1}$. Let $\mathcal{V}$ be $2n + 1 = m$ dimensional, with a non-degenerate symmetric bilinear form (x, y) . Let $\mathfrak{L}$ be the Lie algebra of matrices A which are skew symmetric in relation to (x, y) , i.e. $(xA, y) = -(x, yA)$. Let $\{u_1, \dots, u_m\}$ be a basis for $\mathcal{V}$, and denote the Gram matrix by

$$S = (\sigma_{ij}),$$

$$(u_i, u_j) = \sigma_{ij}.$$

A matrix A is an element of $\mathfrak{L}$ if and only if $(u_i A, u_j) = -(u_i, A u_j)$ for all $1 \leq i, j, m$. We can express $u_i A$ by

$$u_i A = \sum_k \alpha_{ik} u_k,$$

which allows us to rewrite these conditions as

$$\sum_k \alpha_{ik} \sigma_{kj} = - \sum_k \sigma_{ij} \alpha_{jk},$$

$$\begin{aligned}
XS &= -SX^T, \\
X &= (\alpha_{ij}).
\end{aligned} \tag{4.1}$$

As this form is of maximal Witt index, we can choose a basis so that

$$S = \begin{pmatrix} \sigma & 0 & 0 \\ 0 & 0 & \sigma I_n \\ 0 & \sigma I_n & 0 \end{pmatrix},$$

Where σ is some nonzero element of $\mathbb{F}$. We can rescale S so that

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_n \\ 0 & I_n & 0 \end{pmatrix},$$

We can then partition X so that

$$X = \begin{pmatrix} \alpha_{11} & v_1 & v_2 \\ w_1 & Y_{11} & Y_{12} \\ w_2 & Y_{21} & Y_{22} \end{pmatrix},$$

Where $Y_{ij} \in \mathbb{F}^{n \times n}$, and $v_1, v_2, w_1^T, w_2^T \in \mathbb{F}^{1 \times n}$. Then equation 4.1 holds if and only if

$$w_1 = -v_2^T, \tag{4.2}$$

$$w_2 = -v_1^T, \tag{4.3}$$

$$Y_{22} = -Y_{11}^T \tag{4.4}$$

$$Y_{12}^T = -Y_{12}, \tag{4.5}$$

$$Y_{21}^T = -Y_{21}, \tag{4.6}$$

$$\alpha_{11} = 0 \tag{4.7}$$

From these conditions, we get the following set of basis elements for $\mathfrak{L}$,

- Equation 4.4 gives us the elements $H_i = E_{i+1,i+1} - E_{i+n+1,i+n+1}$ and $F_{i,-j} = E_{j+1,i+1} - E_{i+n+1,j+n+1}$ for $i \neq j$,
- Equation 4.6 gives us the elements $F_{i,j} = E_{i+n+1,j+1} - E_{j+n+1,i+1}$ for $i < j$,
- Equation 4.5 gives us the elements $F_{-i,-j} = E_{j+1,i+n+1} - E_{i+1,j+n+1}$ for $i < j$,
- Equation 4.3 Gives us the elements $F_i = E_{1,i+1} - E_{i+n+1,1}$, and
- Equation 4.2 gives us the elements $F_{-i} = E_{i+1,1} - E_{1,i+n+1}$,

for $1 \leq i, j \leq n$.

$-C_n$, or $\mathfrak{sp}_{2n}$. Let $\mathcal{V}$ have dimension $2n = m$, and let (x, y) be a non-degenerate skew bilinear form on $\mathcal{V}$. Let $\mathfrak{L}$ be the Lie algebra of matrices which are skew relative to (x, y) . Choose a basis $\{u_1, \dots, u_m\}$ for $\mathcal{V}$ so that the Gram matrix $Q = ((u_i, u_j))$ is

$$Q = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

We can identify $\mathfrak{L}$ with the Lie algebra of matrices $A \in \mathbb{F}^{m \times m}$ such that $AQ = -QA^T$, which implies that

$$A = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} \tag{4.8}$$

$$X_{ij} \in \mathbb{F}^n \tag{4.9}$$

$$X_{22} = -X_{11}^T, \tag{4.10}$$

$$X_{12}^T = X_{12}, \tag{4.11}$$

$$X_{21}^T = X_{21}. \tag{4.12}$$

These equations allow us to define the following basis:

- From Equation 4.10 we get the elements $H_i = E_{ii} - E_{n+1, n+i}$ and $F_{j, -i} = E_{ij} - E_{j+n, i+n}$ where $i \neq j$,
- From Equation 4.11 we get the elements $F_{-i, -j} = E_{i, j+n} + E_{j, i+n}$ where $i < j$ and $F_{-2i} = E_{i, i+n}$,
- From Equation 4.12 We get $F_{i, j} = E_{i+n, j} + E_{j+n, i}$ where $i < j$ and $F_{2i} = E_{i+n, i}$,

where $1 \leq i, j \leq n$.

$-D_n$, or $\mathfrak{so}_{2n}$. Let $\mathcal{V}$ be $2n = m$ dimensional with a symmetric non-degenerate bilinear form (x, y) of maximal Witt index. Choose a basis $\{u_i\}$ such that the Gram matrix T has the form

$$T = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix},$$

and $\mathfrak{L}$ is expressed as the Lie algebra of matrices A that satisfy $AT = -TA^T$. These are matrices of the form

$$A = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}$$

$$X_{ij} \in \mathbb{F}^n$$

$$X_{22} = -X_{11}^T,$$

$$X_{12}^T = -X_{12},$$

$$X_{21}^T = -X_{21}.$$

These equations allow us to define the following basis:

- $H_i = E_{i, i} - E_{n+i, n+i}$,
- $F_{i, -j} = E_{j, i} - E_{i+n, j+n}$, for $i \neq j$,
- $F_{i, j} = E_{i+n, j} - E_{j+n, i}$ for $i < j$, and

- $F_{-i,-j} = E_{j,i+n} - E_{i,j+n}$ for $i < j$,

where $1 \leq i, j \leq n$.

For the sake of brevity will not include the proofs for the five exceptional Lie algebras. Empirically, these Lie algebras have not been present in the derivations we have computed. Future work may be done to study these cases to determine their likelihood that their elements occur as derivations of adjacency tensors for hypergraphs. $\square$

4.3 Derivations and their associated hypergraph structure

For the purposes of this section, the hypergraph $\mathcal{H}$ is connected, 3-uniform on m vertices, and all derivations are of the form $(0, D, D, D)$. This allows us to embed our derivations into $M_m(\mathbb{F})$, and utilize the results from the previous section. The proof for Theorem 4.3.3 is split into two parts. Subsection 4.3.1 will prove the case of matrices D with support less than three, while subsection 4.3.2 will prove the case of matrices D with support equal to exactly three.

4.3.1 Support less than three

We will first consider all possible matrices of support less than three. We will see which of these matrices can be used to generate derivations of the form $(0, D, D, D)$, and what hypergraph structure they uncover. Images of the hypergraph structures being described are presented in Appendix A.

Proposition 4.3.1. Let $\overline{D} = (0, D, D, D)$ be a derivation of the adjacency tensor of $\mathcal{H}$, where $\mathcal{H}$ is 3-uniform and connected. Let $1 \leq x, y \leq m$, and let a, b, c, d, x , and y be distinct. If the support of D is less than 3, then the following is true

1. The support of D is not equal to 1,
2. D is of the form $\alpha E_{ij} + \beta E_{kl}$ for some $1 \leq i, j, k, l \leq m$, where $\alpha + \beta = 0$,
3. $D = \alpha E_{ab} + \beta E_{cb}$ if and only if

- $\{a, b, c\}, \{a, b, x\} \notin \mathcal{H}$ and
- $(\{a, x, y\} \in \mathcal{H} \iff \{c, x, y\} \in \mathcal{H})$,

4. $D = \alpha E_{ab} + \beta E_{cd}$ if and only if

- $\{a, x, y\} \in \mathcal{H}$ if and only if $x = d$ and $\{b, c, y\} \in \mathcal{H}$
- $\{c, x, y\} \in \mathcal{H}$ if and only if $x = b$ and $\{a, d, y\} \in \mathcal{H}$,

5. $D = \alpha E_{aa} + \beta E_{ba}$ if and only if

- $\{a, b, x\} \notin \mathcal{H}$ and
- $\{a, x, y\} \in \mathcal{H} \iff \{b, x, y\} \in \mathcal{H}$,

6. $D = \alpha E_{aa} + \beta E_{bb}$ if and only if for every $x, y \notin \{a, b\}$,

- $\{a, x, y\} \notin \mathcal{H}$,
- $\{b, x, y\} \notin \mathcal{H}$,

Proof. As everything is finite dimensional, this lemma can be proven by exhausting all possible calculations of the derivation equations. The proofs for 1 and 2 will follow from explicit calculations shown. Recall that $(0, D, D, D)$ is a derivation if and only if it satisfies the following relation on all $u, v, w \in V$

$$\Gamma(Du, v, w) + \Gamma(u, Dv, w) + \Gamma(u, v, Dw) = 0.$$

This can be further simplified by asserting that the relation holds on a basis of e_i, e_j, e_k . Let $1 \leq a, b, c, d, x, y \leq m$ be distinct. Figure 4.1 lists all matrices of support equal to one or two.

The action of αE_{ab} is as follows

$$\alpha E_{ab} e_i = \begin{cases} \alpha e_a & i = b \\ 0 & \text{else} \end{cases}$$

$$\begin{array}{lll}
\alpha E_{ab}, & \alpha E_{aa}, & \alpha E_{ab} + \beta E_{ac}, \\
\alpha E_{ab} + \beta E_{cb}, & \alpha E_{ab} + \beta E_{cd}, & \alpha E_{ab} + \beta E_{ba}, \\
\alpha E_{ab} + \beta E_{ca}, & \alpha E_{aa} + \beta E_{ab}, & \alpha E_{aa} + \beta E_{ba}, \\
\alpha E_{aa} + \beta E_{bc}, & \alpha E_{aa} + \beta E_{bb}. &
\end{array}$$

Figure 4.1: Matrices of support equal to one or two

Since e_i is associated to i we will for simplicity evaluate our tensor on (i, j, k) instead of (e_i, e_j, e_k) . All possible combinations of basis elements to check are as follows

$$\begin{array}{llllll}
(a,b,c) & (a,b,d) & (a,c,d) & (b,c,d) & (a,b,x) & (a,c,x) \\
(a,d,x) & (b,c,x) & (b,d,x) & (c,d,x) & (a,x,y) & (b,x,y) \\
(c,x,y) & (d,x,y) & (a,a,b) & (a,a,c) & (a,a,d) & (b,b,a) \\
(b,b,c) & (b,b,d) & (c,c,a) & (c,c,b) & (c,c,d) & (d,d,a) \\
(d,d,b) & (d,d,c) & (a,a,x) & (b,b,x) & (c,c,x) & (d,d,x)
\end{array}$$

Figure 4.2: Potentially non-trivial triples of basis elements

Certain properties of triples of basis elements (u, v, w) may allow them to hold trivially in the calculations for derivations. For example, (u, u, u) would always hold, as $\Gamma(v, u, u) = 0$. We shall do a few of these calculations, and the rest are left as an exercise to the reader.

First, we will consider the matrix $D = \alpha E_{ab}$, and thus the triple $\overline{D} = (0, D, D, D)$. This is a derivation of Γ if and only if

$$\Gamma(\alpha E_{ab}i, j, k) + \Gamma(i, \alpha E_{ab}j, k) + \Gamma(i, j, \alpha E_{ab}k) = 0$$

for all $1 \leq i, j, k \leq m$. First, we know that if none of i, j, k equal to b , then this relation holds trivially. Therefore, we can consider the case where at least one of i, j, k equals to b . Furthermore, we know that if all three equal to b then $\Gamma(\alpha E_{ab}b, b, b) = \alpha \Gamma(a, b, b) = 0$, and likewise for the other two terms. All other calculations are as follows:

$$(b, x, y) : \Gamma(\alpha E_{ab}b, x, y) + \Gamma(b, \alpha E_{ab}x, y) + \Gamma(b, x, \alpha E_{ab}y)$$

$$\begin{aligned}
&= \alpha\Gamma(a, x, y) \\
(b, b, x) : & \Gamma(\alpha E_{ab}b, b, x) + \Gamma(b, \alpha E_{ab}b, x) + \Gamma(b, b, \alpha E_{ab}x) \\
&= 2\alpha\Gamma(a, b, y) \\
(a, b, x) : & \Gamma(\alpha E_{ab}a, b, x) + \Gamma(a, \alpha E_{ab}b, x) + \Gamma(a, b, \alpha E_{ab}x) \\
&= \alpha\Gamma(a, a, x) \\
(a, b, b) : & \Gamma(\alpha E_{ab}a, b, b) + \Gamma(a, \alpha E_{ab}b, b) + \Gamma(a, b, \alpha E_{ab}b) \\
&= 2\alpha\Gamma(a, a, b) \\
(a, a, b) : & \Gamma(\alpha E_{ab}a, a, b) + \Gamma(a, \alpha E_{ab}a, b) + \Gamma(a, b, \alpha E_{ab}b) \\
&= \alpha\Gamma(a, a, a)
\end{aligned}$$

Now, $\overline{D}$ is a derivation if and only if all of the preceding terms equal to zero. $\Gamma(a, a, b) = 0$ and $\Gamma(a, a, a) = 0$ by defintiiion. $\Gamma(a, x, y) = 0$ if and only if $\{a, x, y\} \notin \mathcal{H}$, and $\Gamma(a, b, x) = 0$ if and only if $\{a, b, x\} \notin \mathcal{H}$. These two conditions are equivalent to saying that $(0, D, D, D)$ is a derivation if and only if a is not a vertex in any hyperedge in $\mathcal{H}$, which contradicts the assumption that $\mathcal{H}$ is 3-uniform and connected. Therefore, $(0, \alpha E_{ab}, \alpha E_{ab}, \alpha E_{ab})$ cannot be a derivation of Γ . The proof for 1 follows from doing this calculation on αE_{aa} .

We will now prove 3, which looks at matrices matrix $D = \alpha E_{ab} + \beta E_{cb}$, and the tuple $(0, D, D, D)$. This is a derivation if and only if

$$\Gamma((\alpha E_{ab} + \beta E_{cb})i, j, k) + \Gamma(i, (\alpha E_{ab} + \beta E_{cb})j, k) + \Gamma(i, j, (\alpha E_{ab} + \beta E_{cb})k) = 0.$$

for all $1 \leq i, j, k \leq m$. The calculations are as follows:

$$\begin{aligned}
(b, x, y) : & \alpha\Gamma(a, x, y) + \beta\Gamma(c, x, y) \\
(a, b, x) : & \alpha\Gamma(a, a, x) + \beta\Gamma(a, c, x) \\
&= \beta\Gamma(a, c, x)
\end{aligned}$$

$$\begin{aligned}
(b, c, x) : & \alpha\Gamma(a, c, x) + \beta\Gamma(c, c, x) \\
& = \alpha\Gamma(a, c, x) \\
(a, b, c) : & \alpha\Gamma(a, a, c) + \beta\Gamma(a, c, c) = 0 \\
(b, b, x) : & 2\alpha\Gamma(a, b, x) + 2\beta\Gamma(b, c, x) \\
(a, b, b) : & 2\beta\Gamma(a, b, c) \\
(b, b, c) : & 2\alpha\Gamma(a, b, c)
\end{aligned}$$

$\overline{D}$ is a derivation if and only if all of the preceding terms equal to zero. $\Gamma(a, b, c) = 0$ and $\beta\Gamma(a, c, x) = 0$ if and only if $\{a, c, x\} \notin \mathcal{H}$ for all x , including $x = b$, i.e. there is no hyperedge $e \in \mathcal{H}$ that contains both a and c as a vertex. Furthermore, we have the relations $\alpha\Gamma(a, x, y) + \beta\Gamma(c, x, y) = 0$ and $2\alpha\Gamma(a, b, x) + 2\beta\Gamma(b, c, x) = 0$. If $\alpha + \beta \neq 0$, then both of these relations require that $\Gamma(a, x, y) = 0$, and $\Gamma(a, b, x) = 0$, which holds if and only if a is not a vertex in any hyperedge in $\mathcal{H}$, a contradiction. Therefore, we must have that $\alpha + \beta = 0$. It follows that $\alpha\Gamma(a, x, y) + \beta\Gamma(c, x, y) = 0$ if and only if $\Gamma(a, x, y) = \Gamma(c, x, y)$, and likewise $2\alpha\Gamma(a, b, x) + 2\beta\Gamma(b, c, x) = 0$ if and only if $\Gamma(a, b, x) = \Gamma(b, c, x)$. This holds if and only if for all $1 \leq x, y \leq m$, where x may equal to b ,

$$\{a, x, y\} \in \mathcal{H} \iff \{c, x, y\} \in \mathcal{H}$$

This concludes the proof for 3. In this proof, we have part of the proof for 2. The rest of the proof for 2 follows from the proofs for 4-6, in which the relation $\alpha + \beta \neq 0$ implies a vertex is isolated. We leave the proofs for 4-6 as an exercise for the reader, as they follow an identical format to the proofs for 1 and 3. $\square$

Example 4.3.2. Consider the hypergraph given by hyperedges

The adjacency tensor of this hypergraph has the following as derivations

$$(0, D, D, D)$$

$\{1, 2, 3\} \quad \{1, 4, 5\} \quad \{1, 3, 8\} \quad \{2, 3, 9\} \quad \{2, 6, 7\} \quad \{3, 8, 9\} \quad \{6, 7, 8\} \quad \{6, 7, 9\}$

Figure 4.3: Example Hypergraph

$$D = E_{1,1} - E_{9,1}$$

$$(0, F, F, F)$$

$$F = E_{2b} - E_{8b}, \quad 1 \leq b \leq 9, \quad b \neq 2, 8$$

4.3.2 Support Less Than or Equal to Exactly Three

Now that we have characterized all matrices of support equal to one and two, we begin characterizing all matrices of support equal to three. Following this, we will have a full characterization of derivations and their associated hypergraph structure. To begin, there are 21 generic matrices of support equal to 3, listed in Table 4.3.2, and 79 possible inputs listed in Table 4.5.

With this information, we can say what each matrix present in the derivation algebra tells us about the hypergraph. For simplicity, we once again assume that all hypergraphs are connected and 3-uniform. Much of the work presented in this section is similar in practice to the proof for Proposition 4.3.1. We separate our cases by the numbered matrix forms as listed in Table 4.3.2

Theorem 4.3.3. Let $\mathfrak{g}$ be the sub-Lie algebra of derivations of the adjacency tensor Γ of a 3-uniform connected hypergraph, where $\mathfrak{g}$ is generated by elements of the form $(0, E, E, E)$. Let $(0, D, D, D)$ be such that D has support less than or equal to three. Then

1. $D = \alpha E_{ab} + \beta E_{cb}$ if and only if

- $\{a, b, c\}, \{a, b, x\} \notin \mathcal{H}$ and
- $(\{a, x, y\} \in \mathcal{H} \iff \{c, x, y\} \in \mathcal{H})$,

2. $D = \alpha E_{ab} + \beta E_{cd}$ if and only if

- $\{a, x, y\} \in \mathcal{H}$ if and only if $x = d$ and $\{b, c, y\} \in \mathcal{H}$

Case	Matrix Forms
1. All in one row	$M1 = \alpha E_{ab} + \beta E_{ac} + \gamma E_{ad}$ $M2 = \alpha E_{aa} + \beta E_{ab} + \gamma E_{ac}$
2. All in one column	$M3 = \alpha E_{ab} + \beta E_{cb} + \gamma E_{db}$ $M4 = \alpha E_{aa} + \beta E_{ba} + \gamma E_{ca}$
3. Two in one row, one elsewhere	$M5 = \alpha E_{ab} + \beta E_{ac} + \gamma E_{de}$ $M6 = \alpha E_{aa} + \beta E_{ab} + \gamma E_{cd}$ $M7 = \alpha E_{ab} + \beta E_{ac} + \gamma E_{dd}$ $M8 = \alpha E_{aa} + \beta E_{ab} + \gamma E_{cc}$
4. Two in one column, one elsewhere	$M9 = \alpha E_{ab} + \beta E_{cb} + \gamma E_{de}$ $M10 = \alpha E_{aa} + \beta E_{ba} + \gamma E_{cd}$ $M11 = \alpha E_{ab} + \beta E_{cb} + \gamma E_{dd}$ $M12 = \alpha E_{aa} + \beta E_{ba} + \gamma E_{cc}$
5. L-shape	$M13 = \alpha E_{ab} + \beta E_{ac} + \gamma E_{db}$ $M14 = \alpha E_{aa} + \beta E_{ab} + \gamma E_{cb}$ $M15 = \alpha E_{ab} + \beta E_{ac} + \gamma E_{bb}$ $M16 = \alpha E_{ab} + \beta E_{aa} + \gamma E_{bb}$ $M17 = \alpha E_{ab} + \beta E_{aa} + \gamma E_{ca}$
6. All rows and columns distinct	$M18 = \alpha E_{ab} + \beta E_{cd} + \gamma E_{ef}$ $M19 = \alpha E_{aa} + \beta E_{bc} + \gamma E_{de}$ $M20 = \alpha E_{aa} + \beta E_{bb} + \gamma E_{cd}$ $M21 = \alpha E_{aa} + \beta E_{bb} + \gamma E_{cc}$

Figure 4.4: All matrices of support equal to 3

(a,a,b)	(a,a,c)	(a,a,d)	(a,a,e)	(a,a,f)	(b,b,a)
(b,b,c)	(b,b,d)	(b,b,e)	(b,b,f)	(c,c,a)	(c,c,b)
(c,c,d)	(c,c,e)	(c,c,f)	(d,d,a)	(d,d,b)	(d,d,c)
(d,d,d)	(d,d,e)	(d,d,f)	(e,e,a)	(e,e,b)	(e,e,c)
(e,e,d)	(e,e,e)	(e,e,f)	(f,f,a)	(f,f,b)	(f,f,c)
(f,f,d)	(f,f,e)	(c,e,f)	(d,e,f)	(c,d,e)	(c,d,f)
(a,b,c)	(a,b,d)	(a,b,e)	(a,b,f)	(a,c,d)	(a,c,e)
(a,c,f)	(a,d,e)	(a,d,f)	(a,e,f)	(b,c,d)	(b,c,e)
(b,c,f)	(b,d,e)	(b,d,f)	(b,e,f)	(d,e,x)	(d,f,x)
(a,a,x)	(b,b,x)	(c,c,x)	(d,d,x)	(e,e,x)	(f,f,x)
(a,b,x)	(a,c,x)	(a,d,x)	(a,e,x)	(a,f,x)	(b,c,x)
(b,d,x)	(b,e,x)	(b,f,x)	(c,d,x)	(c,e,x)	(c,f,x)
(e,f,x)	(a,x,y)	(b,x,y)	(c,x,y)	(d,x,y)	(e,x,y)
(f,x,y)					

Figure 4.5: All inputs for Matrices of Support Equal to Three

- $\{c, x, y\} \in \mathcal{H}$ if and only if $x = b$ and $\{a, d, y\} \in \mathcal{H}$,

3. $D = \alpha E_{aa} + \beta E_{ba}$ if and only if

- $\{a, b, x\} \notin \mathcal{H}$, and
- $\{a, x, y\} \in \mathcal{H} \iff \{b, x, y\} \in \mathcal{H}$,

4. $D = \alpha E_{aa} + \beta E_{bb}$ if and only if for every $x, y \notin \{a, b\}$,

- $\{a, x, y\} \notin \mathcal{H}$, and
- $\{b, x, y\} \notin \mathcal{H}$,

5. $D = \alpha E_{aa} + \beta E_{ab} + \gamma E_{cc}$ if and only if

- $\alpha + \gamma = 0$,
- $a \in e \in \mathcal{H}$ implies that $e = \{a, c, x\}$ for some $x \in V - \{a, c\}$, and
- $c \in e \in \mathcal{H}$ implies that $e = \{a, c, x\}$ for some $x \in V - \{a, c\}$

6. $D = \alpha E_{aa} + \beta E_{bb} + \gamma E_{cd}$ if and only if

- $\alpha + \beta = 0$,
- $a \in e \in \mathcal{H}$ implies that $e = \{a, b, x\}$ for some $x \in V - \{a, b\}$, and
- $b \in e \in \mathcal{H}$ implies that $e = \{a, b, x\}$ for some $x \in V - \{a, b\}$

7. $D = \alpha E_{aa} + \beta E_{bb} + \gamma E_{cc}$ where $(\alpha + \beta + \gamma) = 0$ if and only if

- $\mathcal{H} = \{\{a, b, c\}\}$
- $(\alpha + \beta) \neq 0$, $(\alpha + \gamma) \neq 0$, and $(\beta + \gamma) \neq 0$,

8. $D = \alpha E_{ab} + \beta E_{ac} + \gamma E_{de}$ if and only if

- $\{a, e, x\}, \{b, d, x\}, \{c, d, x\} \in \mathcal{H}$ for some $x \in V - \{a, e\}$,
- $\{b, c, d\}, \{d, x, y\} \notin \mathcal{H}$ for all $x, y \in \mathcal{H}$,
- $\alpha + \gamma = 0$, and

- $\beta + \gamma = 0$

9. $D = \alpha E_{ab} + \beta E_{ac} + \gamma E_{db}$ if and only if

- $\alpha + \gamma = 0$,
- $\beta + \gamma = 0$,
- $\{a, b, c\} \in \mathcal{H}$ for $x \notin \{c, d\}$ if and only if $\{b, d, x\} \in \mathcal{H}$,
- $\{c, d, x\} \in \mathcal{H}$ for $x \notin \{a, b\}$ if and only if $\{a, b, c\} \in \mathcal{H}$, and
- All other hyperedges in $\mathcal{H}$ are of the form $\{b, c, x\}$, $\{b, x, y\}$, $\{c, x, y\}$ and $\{x, y, z\}$.

10. $D = \alpha E_{ab} + \beta E_{cb} + \gamma E_{db}$ if and only if

- $\{a, x, y\}, \{c, x, y\}, \{d, x, y\} \in \mathcal{H}$,
- $\{a, c, x\} \notin \mathcal{H}, \{a, d, x\} \notin \mathcal{H}, \{c, d, x\} \notin \mathcal{H}$ for $x \in \mathcal{H} - \{a, c, d\}$,
- $\{a, b, c\}, \{a, c, d\}, \{a, b, d\}, \{b, c, d\} \notin \mathcal{H}$, and
- $\alpha + \beta + \gamma = 0$.

11. $D = \alpha E_{ab} + \beta E_{cd} + \gamma E_{ef}$, where $\alpha + \beta + \gamma \neq 0$, if and only if

- $\mathcal{H} = \{\{a, d, f\}, \{b, c, f\}, \{b, d, e\}\}$

12. $D = \alpha E_{aa} + \beta E_{ba} + \gamma E_{ca}$ if and only if

- $\{a, b, c\}, \{a, b, x\}, \{a, c, x\}, \{b, c, x\} \notin \mathcal{H}$ for $x \in V - \{a, b, c\}$,
- $\{a, x, y\}, \{b, x, y\}, \{c, x, y\} \in \mathcal{H}$ for $x, y \in V - \{a, b, c\}$, and
- $(\alpha + \beta + \gamma) = 0$.

13. $D = \alpha E_{ab} + \beta E_{cb} + \gamma E_{de}$, where $\alpha + \beta \neq 0$, if and only if

- $\{a, b, d\}, \{b, c, d\}, \{a, b, x\}, \{b, c, x\}, \{a, b, e\}, \{b, c, e\}, \{a, x, y\}, \{c, x, y\}, \{a, c, e\} \notin \mathcal{H}$,

- $\{b, d, x\}, \{a, e, x\}, \{c, e, x\} \in \mathcal{H}$, and
 - $\alpha + \beta + \gamma = 0$
14. $D = \alpha E_{ab} + \beta E_{cb} + \gamma E_{de}$, where $\alpha + \beta = 0$, $\{a, b, d\} \notin \mathcal{H}$, $\alpha + \gamma \neq 0$, and $\beta + \gamma = 0$, if and only if
- $\{b, d, x\}, \{c, e, x\} \in \mathcal{H}$,
 - $\{a, e, x\} \notin \mathcal{H}$,
15. $D = \alpha E_{ab} + \beta E_{cb} + \gamma E_{de}$, where $\alpha + \beta = 0$, $\{a, b, d\} \notin \mathcal{H}$, $\alpha + \gamma = 0$, and $\beta + \gamma \neq 0$, if and only if
- $\{b, d, x\}, \{a, e, x\} \in \mathcal{H}$,
 - $\{c, e, x\} \notin \mathcal{H}$,
16. The support of D is not equal to 1,
17. If D is of the form $\alpha E_{ij} + \beta E_{kl}$ for some $1 \leq i, j, k, l \leq m$, then $\alpha + \beta = 0$,

Furthermore, D can only be one of the previously listed matrices. Additionally $\mathfrak{g}$ is generated by $(0, D, D, D)$ where D has support less than or equal to three.

For each of the 18 cases, we provide images for the described hyperedges in the Appendix. They will be referenced as Case n. For example, $D = \alpha E_{ab} + \beta E_{cd}$ is shown in Figure A.1

Proof. The first four cases are outlined in Proposition 4.3.1. For all matrices of support equal to three, we reference Table 4.3.2 for the cases we must check. For each case, we outline either why it can never be a derivation, or what hyperedges must exist for it to be a derivation. Before we evaluate our cases, we note a common contradiction which may occur. All of the support three derivations are of the form $\alpha E_{ab} + \beta E_{cd} + \gamma E_{ef}$, where α , β , and γ are all nonzero. If in our calculations we get an implication that $(\alpha + \beta) = (\alpha + \gamma) = (\beta + \gamma) = 0$, this would imply that at least one of α , β , or γ is zero, a contradiction.

Cases M1, M2, M6, M7, M10, M11, M12, M14, M16, and M17: These derivations would exist if and only if there is an isolated vertex. To see why this is the case, we look at the calculations for $M1$, as the rest follow similarly. The derivation equation must hold for the basis elements (b,b,c) . By inputting these entries into our derivation equation, we get

$$2\alpha\Gamma(a, b, c),$$

Therefore, $(0, M1, M1, M1)$ is a derivation only if $\Gamma(a, b, c) = 0$. As Γ is an adjacency tensor, this holds if and only if $\{a, b, c\} \notin \mathcal{H}$. Continuing this with different triples (i, j, k) , we eventually get that $(0, M1, M1, M1)$ is a derivation if and only if a is an isolated vertex, which violates our assumption that $\mathcal{H}$ is 3-uniform and connected.

Case M8: Such a derivation exists if and only if $\alpha + \gamma = 0$ and a and c are only in hyperedges of the form $\{a, c, x\}$, for arbitrary vertices $x \in V - \{a, c\}$.

Case M20: Such a derivation would exist if and only $\alpha + \beta = 0$ and for all hyperedges with either a or b as elements, such hyperedges are of the form $\{a, b, x\}$ for some $x \in V - \{a, b\}$

Case M21: $(0, M21, M21, M21)$ is a derivation only if the following hold

$$(\alpha + \beta + \gamma)\Gamma(a, b, c) = 0,$$

$$(\alpha + \beta)\Gamma(a, b, x) = 0,$$

$$(\alpha + \gamma)\Gamma(a, c, x) = 0,$$

$$(\beta + \gamma)\Gamma(b, c, x) = 0,$$

$$\Gamma(a, x, y) = 0,$$

$$\Gamma(b, x, y) = 0,$$

$$\Gamma(c, x, y) = 0,$$

First, we can say definitively that any derivation of the form of $M21$ would arise from a hypergraph where the vertices a , b , and c were only in hyperedges containing at least two of the three. We now go through the cases of what the sum of the coefficients may equal to.

Case M21 (a): If $(\alpha + \beta + \gamma) \neq 0$, and any one of $(\alpha + \beta)$, $(\alpha + \gamma)$, or $(\beta + \gamma)$ is not equal to zero, then that would imply that there is an isolated vertex, which is a contradiction. Therefore, $(\alpha + \beta + \gamma) \neq 0$ implies that $(\alpha + \beta) = (\alpha + \gamma) = (\beta + \gamma) = 0$, which implies that one of α , β , or γ equals to zero, a contradiction.

Case M21 (b): $(\alpha + \beta + \gamma) = 0$, in which case $\{a, b, c\}$ may be a hyperedge. From here, we have eight cases. Due to symmetry we only need to consider two.

- i If any one of $(\alpha + \beta)$, $(\alpha + \gamma)$, or $(\beta + \gamma) = 0$, then one of α , β , or γ equals to zero, a contradiction to $M21$ having support equal to three.
- ii If $(\alpha + \beta) \neq 0$, $(\alpha + \gamma) \neq 0$, and $(\beta + \gamma) \neq 0$, then this would imply that $\{a, b, c\}$ is the only hyperedge that could contain a , b , or c . For this to not contradict the assumption that $\mathcal{H}$ is connected, we must have that $\mathcal{H} = \{\{a, b, c\}\}$. Furthermore, this implies that the only equation which must be satisfied for the derivations of the associated adjacency tensor is

$$\Gamma(Da, b, c) + \Gamma(a, Db, c) + \Gamma(a, b, Dc) = 0$$

and $M21$ is exactly

$$\begin{bmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{bmatrix}$$

Case M5: $\{a, e, x\} \in \mathcal{H}$ for some $x \in V$, otherwise a would be isolated. This then implies that $\{a, e, x\} \in \mathcal{H}$ if and only if

- $\{b, d, x\}, \{c, d, x\} \in \mathcal{H}$,

- $\alpha + \gamma = 0$,
- $\beta + \gamma = 0$, and
- $\{b, c, d\}, \{d, x, y\} \notin \mathcal{H}$

all hold.

Case M13: If $\alpha + \gamma \neq 0$, then $M13$ is a derivation only if $\Gamma(a, b, x) = \Gamma(b, d, x) = 0$, which holds if and only if a is an isolated vertex, a contradiction. Therefore, $\alpha + \gamma = 0$. Similarly, $\beta + \gamma = 0$. From here, we have that $M13$ is a derivation if and only if for $x, y, z \notin \{a, b, c, d\}$

- $\{a, b, x\} \in \mathcal{H}$ for $x \notin \{c, d\}$, if and only if $\{b, d, x\} \in \mathcal{H}$,
- $\{c, d, x\} \in \mathcal{H}$ for $x \notin \{a, b\}$ if and only if $\{a, b, x\} \in \mathcal{H}$, and
- All other hyperedges are of the form $\{b, c, x\}$, $\{b, x, y\}$, $\{c, x, y\}$, and $\{x, y, z\}$

all hold.

Cases M15 and M19: $M15$ and $M19$ are derivations only if a is an isolated vertex, a contradiction.

Case M18: From relations such as $\beta\Gamma(a, c, f) + \gamma\Gamma(a, d, e)$, we have that $M18$ is a derivation only if

$$\{a, c, f\} \in \mathcal{H} \iff \{a, d, e\} \in \mathcal{H} \iff \{b, c, e\} \in \mathcal{H}$$

If $\{a, c, f\} \in \mathcal{H}$, then $\alpha + \gamma = 0$, $\beta + \gamma = 0$, and $\alpha + \beta = 0$, implying that one of α , β , or γ is zero, a contradiction. Therefore,

$$\{a, c, f\} \notin \mathcal{H}, \quad \{a, d, e\} \notin \mathcal{H}, \quad \{b, c, e\} \notin \mathcal{H}.$$

We now look at the relation $\alpha\Gamma(a, d, f) + \beta\Gamma(b, c, f) + \gamma\Gamma(b, d, e)$. We first assume that $\alpha + \beta + \gamma \neq 0$.

Case M18(a):

- i** If $\{a, d, f\}, \{b, c, f\}, \{b, d, e\} \in \mathcal{H}$, then $\alpha\Gamma(a, d, f) + \beta\Gamma(b, c, f) + \gamma\Gamma(b, d, e) \neq 0$, which would mean that $M18$ is not a derivation.
- ii** If only one of $\{a, d, f\}, \{b, c, f\}, \{b, d, e\}$ is an element of $\mathcal{H}$, we would have $\alpha = 0, \beta = 0$, or $\gamma = 0$, a contradiction to $M18$ having support equal to three.
- iii** Assume only two of $\{a, d, f\}, \{b, c, f\}, \{b, d, e\}$ are hyperedges in $\mathcal{H}$. Without loss of generality, assume $\{a, d, f\} \notin \mathcal{H}$. This implies that $\beta + \gamma = 0$, which further implies that $\{c, f, x\} \in \mathcal{H} \iff \{d, e, x\} \in \mathcal{H}$.

We now have two more subcases to consider

Case M18(a)iii.i: Assume that $\alpha + \beta = 0$. This then implies that $\alpha + \gamma \neq 0, \{a, d, x\} \in \mathcal{H} \iff \{b, c, x\} \in \mathcal{H}$, and that $\{a, f, x\}, \{b, e, x\} \notin \mathcal{H}$.

Case M18(a)iii.ii: Assume that $\alpha + \beta \neq 0$ and $\alpha + \gamma \neq 0$. This then implies that

$$\{a, d, x\}, \{b, c, x\}, \{a, f, x\}, \{b, e, x\} \notin \mathcal{H}.$$

However, this implies that a is an isolated vertex, a contradiction

Case M18(b): We now assume that $\alpha + \beta + \gamma = 0$. The remaining expressions to consider are $\alpha\Gamma(a, d, x) + \beta\Gamma(b, c, x)$, $\alpha\Gamma(a, f, x) + \gamma\Gamma(b, e, x)$, and $\beta\Gamma(c, f, x) + \gamma\Gamma(d, e, x)$. Without loss of generality, assume that $\{a, d, x\} \in \mathcal{H}$. This would then imply that $\alpha + \beta = 0$, which would then imply that $\gamma = 0$, a contradiction. Therefore, none of $\{a, d, x\}, \{b, c, x\}, \{a, f, x\}, \{b, e, x\}, \{c, f, x\}$, or $\{d, e, x\}$ may be hyperedges. Therefore, $\{a, d, f\}, \{b, c, f\}$, and $\{b, d, e\}$ are hyperedges that must exist, as otherwise we would have an isolated vertex. Now, for $\mathcal{H}$ to be connected, there can only be these six vertices.

Case M3: First, we have that $\{a, c, d\} \notin \mathcal{H}$. We have equations $\alpha\Gamma(a, b, c) + \Gamma(a, b, d) = 0$, $\alpha\Gamma(a, b, c) + \gamma\Gamma(b, c, d) = 0$ and $\alpha\Gamma(a, b, d) + \beta\Gamma(b, c, d) = 0$. As α, β , and γ are all nonzero, we can rescale these to be $\frac{\alpha^2}{\gamma}\Gamma(a, b, c) + \alpha\Gamma(a, b, d) = 0$, $\frac{\alpha\beta}{\gamma}\Gamma(a, b, c) + \beta\Gamma(b, c, d) = 0$ and

$\alpha\Gamma(a, b, d) + \beta\Gamma(b, c, d) = 0$. This then implies that $\{a, b, c\} \notin \mathcal{H}$, $\{a, b, d\} \notin \mathcal{H}$, and $\{b, c, d\} \notin \mathcal{H}$.

We have two main subcases.

Case M3 (a): If $\alpha + \beta = \alpha + \gamma = \beta + \gamma = 0$, then at least one of α , β , or γ is zero, a contradiction.

Case M3 (b): If any one of $\alpha + \beta$, $\alpha + \gamma$, or $\beta + \gamma$ are nonzero, this implies that

$$\{a, c, x\} \notin \mathcal{H}, \quad \{a, d, x\} \notin \mathcal{H}, \quad \{c, d, x\} \notin \mathcal{H}, \quad x \in \mathcal{H} - \{a, c, d\}.$$

Without loss of generality, assume that $\alpha + \beta \neq 0$. We consider the equation $\alpha\Gamma(a, x, y) + \beta\Gamma(c, x, y) + \gamma\Gamma(d, x, y) = 0$. $\Gamma(a, x, y) = 0$ if and only if $\{a, x, y\} \notin \mathcal{H}$, which would imply that a is an isolated vertex. Similarly, we must have that $\Gamma(c, x, y) = 1$ and $\Gamma(d, x, y) = 1$. This then implies that $\{a, x, y\}, \{c, x, y\}, \{d, x, y\} \in \mathcal{H}$, and $\alpha + \beta + \gamma = 0$.

Case M4: First, $\{a, b, c\} \notin \mathcal{H}$. with a similar calculation to **Case M3**, we also have that $\{a, b, x\} \notin \mathcal{H}$, $\{a, c, x\} \notin \mathcal{H}$, and $\{b, c, x\} \notin \mathcal{H}$. Our last equation to consider in this case is $\alpha\Gamma(a, x, y) + \beta\Gamma(b, x, y) + \gamma\Gamma(c, x, y) = 0$. For none of a , b , or c to be isolated vertices, we must have $\{a, x, y\}, \{b, x, y\}, \{c, x, y\} \in \mathcal{H}$. This implies that $(\alpha + \beta + \gamma) = 0$.

Case M9: We have that

$$\{a, b, c\}, \{a, d, e\}, \{b, d, e\}, \{c, d, e\}, \{d, e, x\}, \{a, c, d\}, \{a, c, x\}, \{a, d, x\}, \{c, d, x\}, \{d, x, y\} \notin \mathcal{H}.$$

This implies that at least one of $\{a, d, b\}$, $\{b, c, d\}$, or $\{b, d, x\}$ is a hyperedge. Furthermore, there is no restriction on $\{b, e, x\}$. Therefore, if it appears as though b or e may be an isolated vertex, we can assume that $\{b, e, x\}$ exists so that a contradiction is not made. We have two subcases

Case M9(a): If $\alpha + \beta \neq 0$, this further implies that $\{a, b, d\}, \{b, c, d\} \notin \mathcal{H}$. This then implies that $\{b, d, x\} \in \mathcal{H}$, and that

$$\{a, b, x\}, \{b, c, x\}, \{a, b, e\}, \{b, c, e\}, \{a, x, y\}, \{c, x, y\}, \{a, c, e\} \notin \mathcal{H}.$$

This implies that $\{a, e, x\} \in \mathcal{H}$, as otherwise a would be an isolated vertex. Similarly $\{c, e, x\} \in \mathcal{H}$. Together, this tells us that $\alpha + \beta + \gamma = 0$.

Case M9(b): If $\alpha + \beta = 0$, then M9 is a derivation only if the following hold

$$\begin{aligned} \{a, b, d\} \in \mathcal{H} &\iff \{b, c, d\} \in \mathcal{H}, \quad \{a, b, e\} \in \mathcal{H} \iff \{b, c, e\} \in \mathcal{H}, \\ \{a, b, x\} \in \mathcal{H} &\iff \{b, c, x\} \in \mathcal{H}, \quad \{a, x, y\} \in \mathcal{H} \iff \{c, x, y\} \in \mathcal{H}, \\ \alpha\Gamma(a, e, x) + \beta\Gamma(c, e, x) + \gamma\Gamma(b, d, x) &= 0. \end{aligned}$$

There are now multiple subcases we must consider, each determined by which hyperedges exist. We first note the following implications

- If $\{b, c, d\} \in \mathcal{H}$ and $\alpha + \gamma \neq 0$, then the first assumption implies that $\{b, c, d\} \in \mathcal{H}$, while the second implies that $\{b, c, d\} \in \mathcal{H}$, a contradiction.
- If $\{a, b, d\} \in \mathcal{H}$ and $\beta + \gamma \neq 0$, then the first assumption implies that $\{a, b, d\} \in \mathcal{H}$, while the second implies that $\{a, b, d\} \in \mathcal{H}$, a contradiction.

Case M9(b)i: If $\alpha + \gamma = 0$, and that $\beta + \gamma = 0$, then $\gamma = 0$, a contradiction.

Case M9(b)ii: Assume that $\{a, b, d\} \notin \mathcal{H}$, $\alpha + \gamma \neq 0$, and that $\beta + \gamma = 0$. Then $\{b, d, x\}, \{c, e, x\} \in \mathcal{H}$, while $\{a, e, x\} \notin \mathcal{H}$

Case M9(b)iii: Assume that $\{a, b, d\} \notin \mathcal{H}$, $\alpha + \gamma = 0$, and that $\beta + \gamma \neq 0$. Then $\{b, d, x\}, \{a, e, x\} \in \mathcal{H}$, while $\{c, e, x\} \notin \mathcal{H}$.

Case M9(b)iv: Assume that $\{a, b, d\} \notin \mathcal{H}$, $\{a, e, x\}, \{c, e, x\} \in \mathcal{H}$, $\alpha + \gamma \neq 0$, and that $\beta + \gamma \neq 0$. Then $\{b, d, x\} \in \mathcal{H}$, and

$$\alpha\Gamma(a, e, x) + \beta\Gamma(c, e, x) + \gamma\Gamma(b, d, x) = \alpha + \beta + \gamma = \gamma,$$

which is equal to zero only if $\gamma = 0$, a contradiction to M9 having support equal to zero.

Case M9(b)v: Assume that $\{a, b, d\} \notin \mathcal{H}$, $\{a, e, x\} \notin \mathcal{H}$, $\{c, e, x\} \in \mathcal{H}$, $\alpha + \gamma \neq 0$, and that $\beta + \gamma \neq 0$. Then

$$\alpha\Gamma(a, e, x) + \beta\Gamma(c, e, x) + \gamma\Gamma(b, d, x) = \beta + \gamma \neq 0,$$

A contradiction to $M19$ being a derivation.

Case M9(b)vi: Assume that $\{a, b, d\} \notin \mathcal{H}$, $\{a, e, x\} \in \mathcal{H}$, $\{c, e, x\} \notin \mathcal{H}$, $\alpha + \gamma \neq 0$, and that $\beta + \gamma \neq 0$. Then

$$\alpha\Gamma(a, e, x) + \beta\Gamma(c, e, x) + \gamma\Gamma(b, d, x) = \alpha + \gamma \neq 0,$$

Case M9(b)vii: Assume that $\{a, b, d\} \notin \mathcal{H}$, $\{a, e, x\}, \{c, e, x\} \notin \mathcal{H}$, $\alpha + \gamma \neq 0$, and that $\beta + \gamma \neq 0$. Then

$$\alpha\Gamma(a, e, x) + \beta\Gamma(c, e, x) + \gamma\Gamma(b, d, x) = \gamma \neq 0,$$

Now, whether or not any one of $\{a, b, e\}$, $\{a, b, x\}$, or $\{a, x, y\}$ are or are not elements of $\mathcal{H}$ does not lead to any further implications other than the existence (or lack there of) of hyperedges, and as such do not lead to any further subcases.

We can therefore simplify and rename our cases for $M9$ as follows:

Case M9(a): If $\alpha + \beta \neq 0$, then

$$\{a, b, d\}, \{b, c, d\}, \{a, b, x\}, \{b, c, x\}, \{a, b, e\}, \{b, c, e\}, \{a, x, y\}, \{c, x, y\}, \{a, c, e\} \notin \mathcal{H}$$

$$\{b, d, x\}, \{a, e, x\}, \{c, e, x\} \in \mathcal{H}$$

$$\alpha + \beta + \gamma = 0$$

Case M9(b): If $\alpha + \beta = 0$, $\{a, b, d\} \notin \mathcal{H}$, $\alpha + \gamma \neq 0$, and $\beta + \gamma = 0$, then

$$\{b, d, x\}, \{c, e, x\} \in \mathcal{H}$$

$$\{a, e, x\} \notin \mathcal{H}$$

Case M9(c): If $\alpha + \beta = 0$, $\{a, b, d\} \notin \mathcal{H}$, $\alpha + \gamma = 0$, and $\beta + \gamma \neq 0$, then

$$\{b, d, x\}, \{a, e, x\} \in \mathcal{H}$$

$$\{c, e, x\} \notin \mathcal{H}$$

We do not need to consider matrices of support greater than three. As outlined in Lemma 4.2.1, these matrices will generate the full Lie algebra.

□

4.3.3 Returning to High School

With Theorem 4.3.3, we can give a more accurate analysis of our High School data set. The reason that the dimension of the derivation algebra of the adjacency tensor was (insert calculation) was that there were certain features present in the hypergraph that led to more interesting derivations exist.

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Appendix A

Hyperedges in Theorem 4.3.3

The following is the list of hyperedges which are implied to exist by the existence of derivations as stated in Theorem 4.3.3:

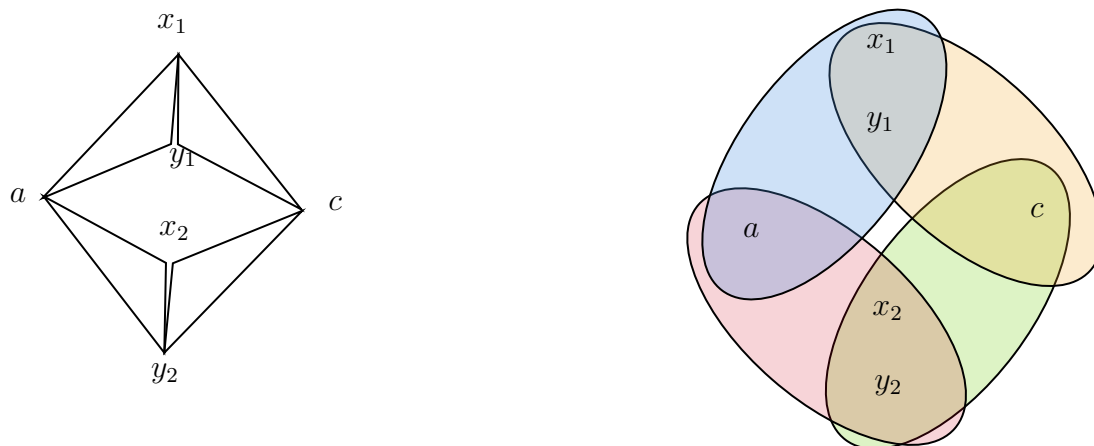


Figure A.1: Theorem 4.3.3: Case 1

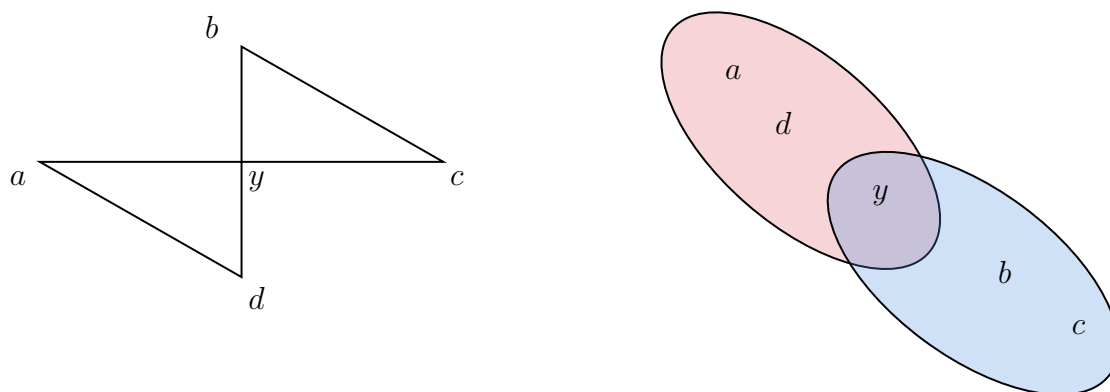


Figure A.2: Theorem 4.3.3: Case 2

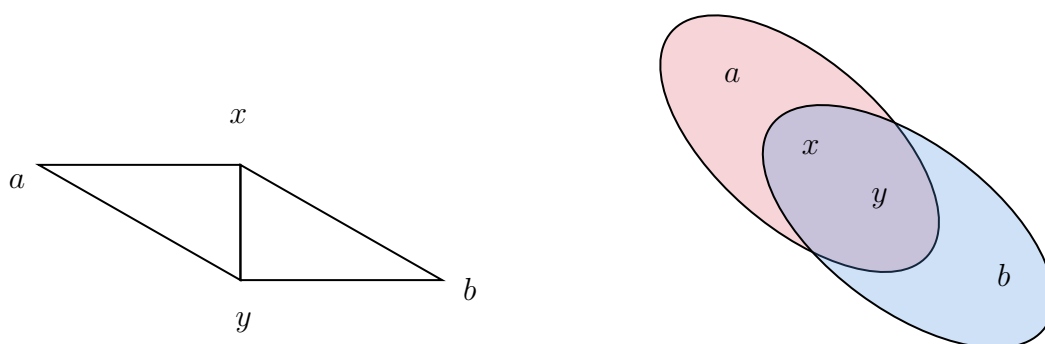


Figure A.3: Theorem 4.3.3: Case 3

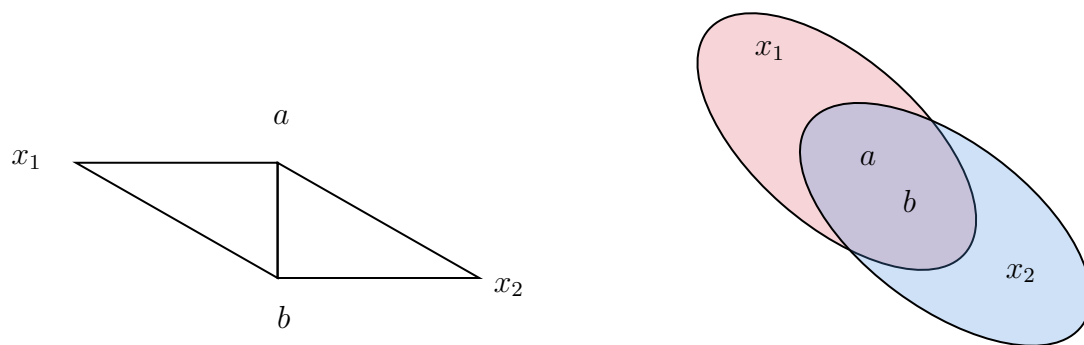


Figure A.4: Theorem 4.3.3: Case 4

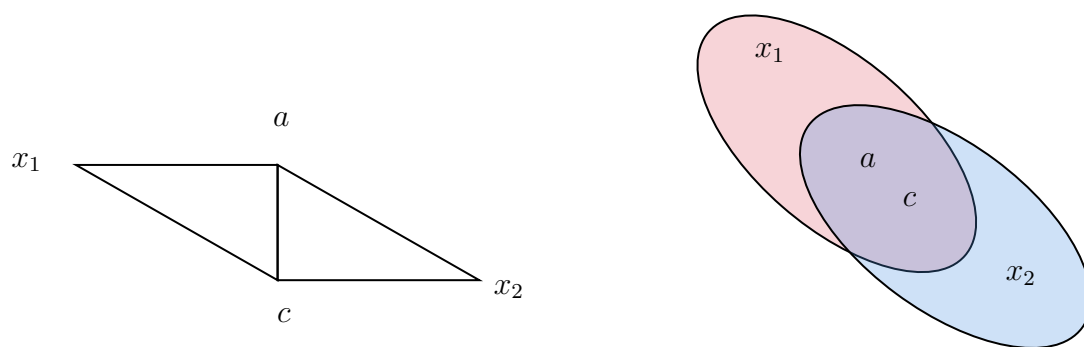


Figure A.5: Theorem 4.3.3: Case 5

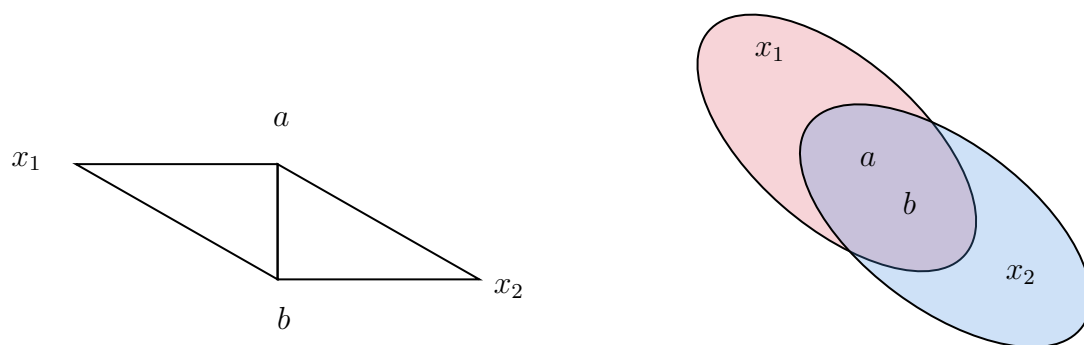


Figure A.6: Theorem 4.3.3: Case 6

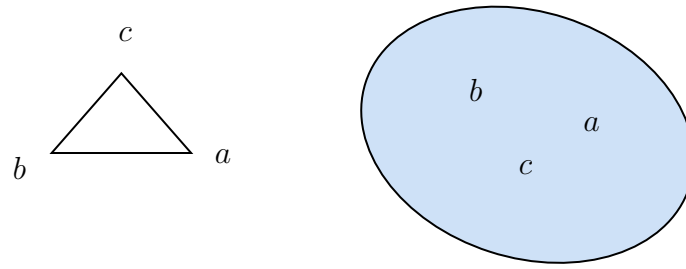


Figure A.7: Theorem 4.3.3: Case 7

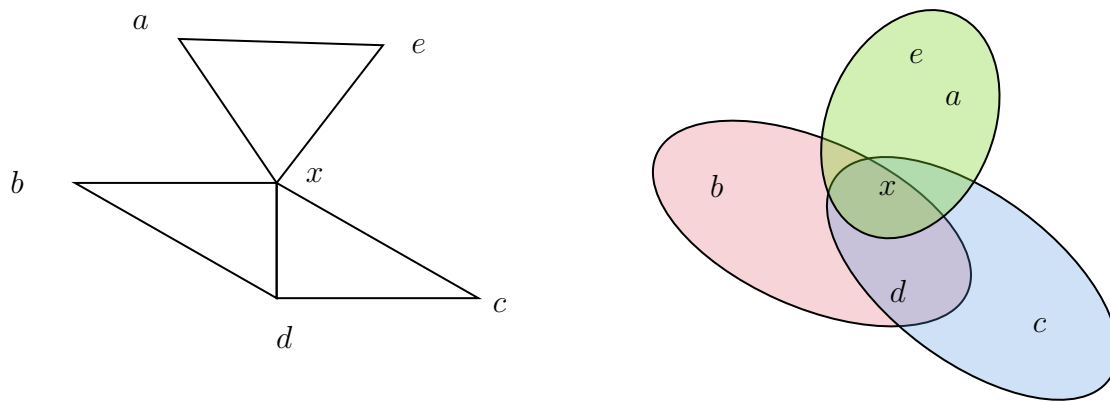


Figure A.8: Theorem 4.3.3: Case 8

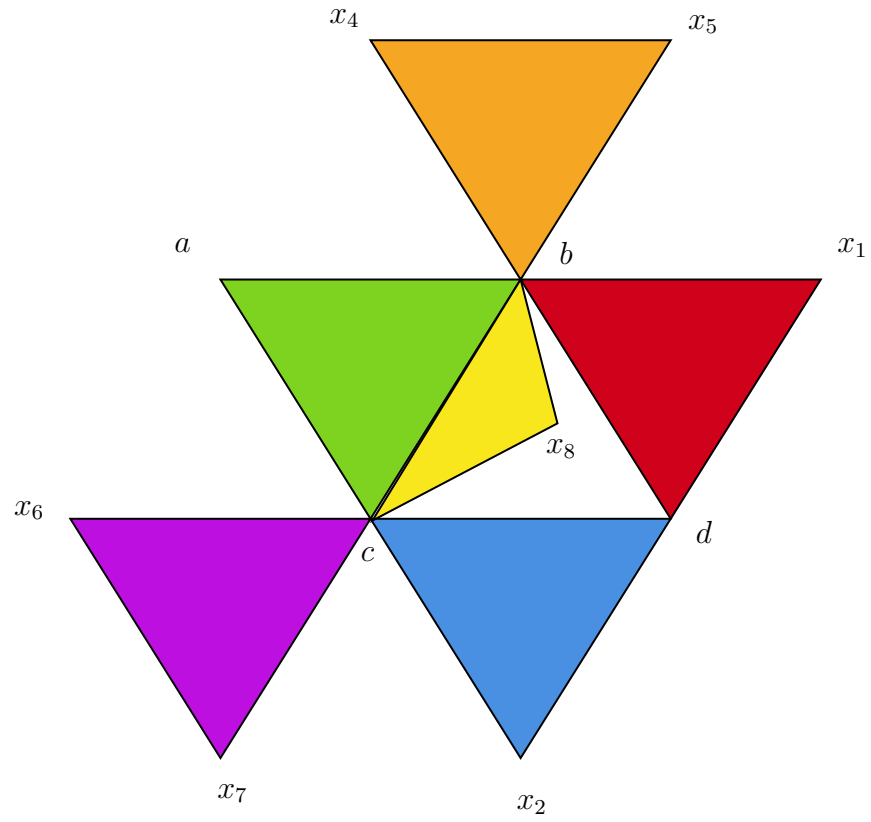


Figure A.9: Theorem 4.3.3: Case 9

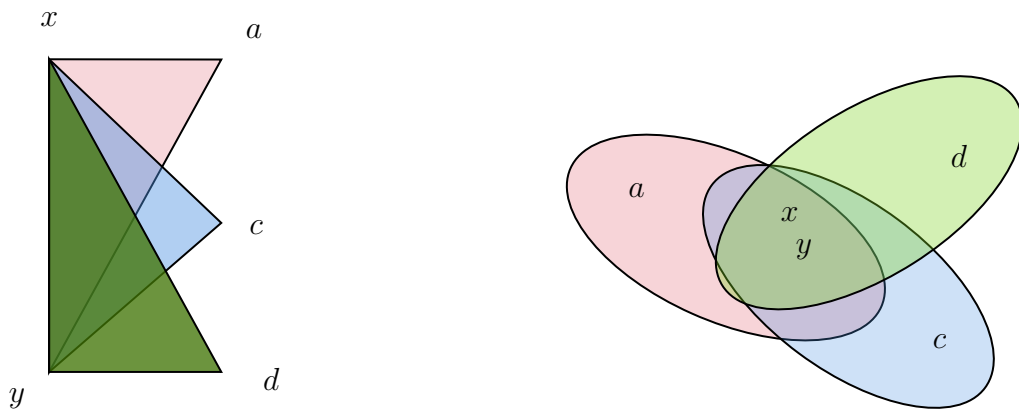


Figure A.10: Theorem 4.3.3: Case 10

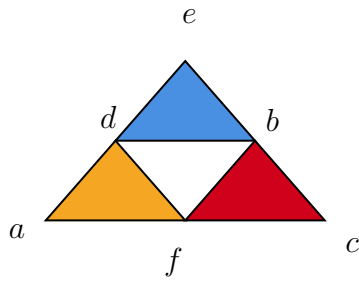


Figure A.11: Theorem 4.3.3: Case 11

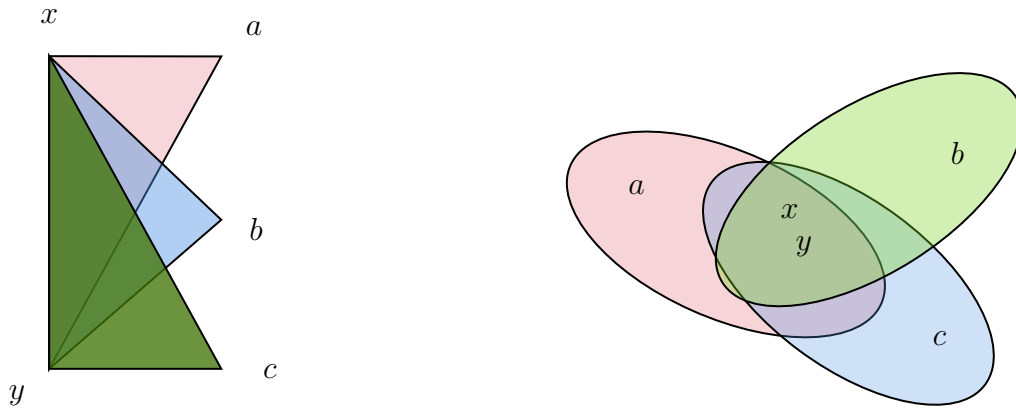


Figure A.12: Theorem 4.3.3: Case 12

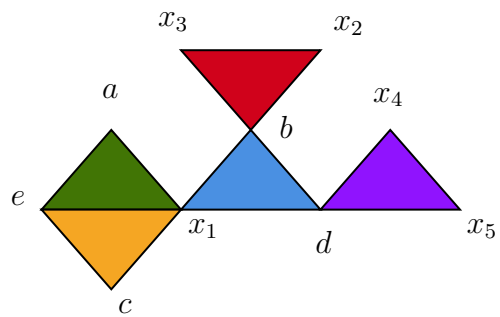


Figure A.13: Theorem 4.3.3: Case 13

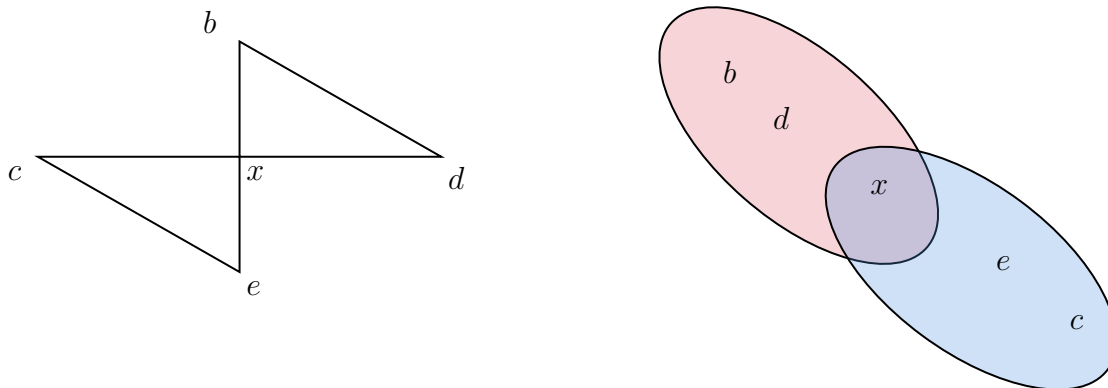


Figure A.14: Theorem 4.3.3: Case 14

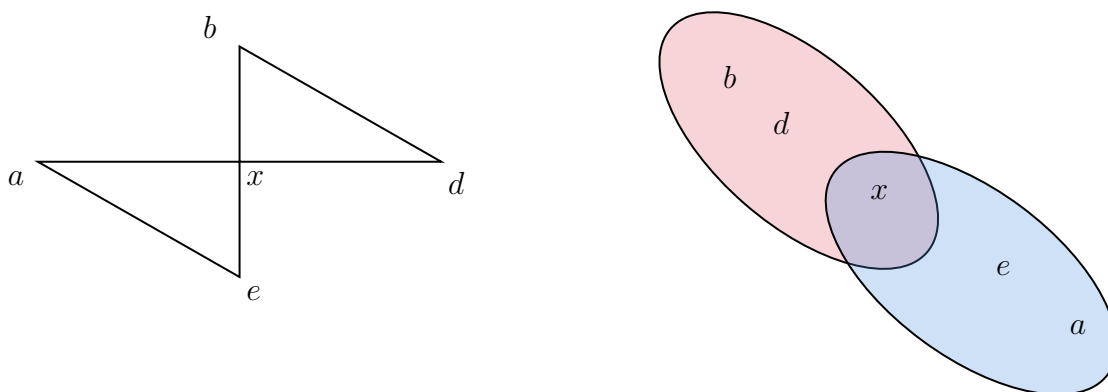


Figure A.15: Theorem 4.3.3: Case 15